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LEARNER'S Pocketbook

2nd Edition

A pocketful of tips, tools
and accelerated learning
techniques to help
anyone involved in the
learning process get
the most from it

Paul Hayden

foreword by **Tony Buzan**

THE LEARNER'S POCKETBOOK

By Paul Hayden

Drawings by Phil Hailstone

“Should prove very useful to those studying for exams, also those with an interest in how we learn. I like the idea of being able to ‘dip into’ different areas as and when needed.”

**Hazel Garvey, Training & Communications Manager, Professional Standards Office,
The Institute of Chartered Accountants**

“The Learner’s Pocketbook provides a treasure chest of ideas and useful information on learning. This is one book that I have revisited time after time and it has provided inspiration to develop our own sessions for managers on effective learning.”

Robbie Lightfoot, HR Development Manager, United Utilities, Contract Solutions

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FOREWORD BY TONY BUZAN

Global business leaders are crying out for more brainpower, the development of **intellectual capital**, and investment in the most powerful currency in the world – the currency of **intelligence**.

This growing development of the awareness *by intelligence of intelligence* and the fact that this quality is not only multiple but can be nurtured and grown, is one of the great beacons of hope for the future of our race. When humankind becomes truly mentally literate – understanding both the alphabet of the neurophysiology of the brain and the alphabet of its behaviour, including memory, creativity, thinking, reading and learning – the world will, in all probability, approach the utopia which for many millennia has been the dream of so many different societies.

Paul Hayden's delightful **The Learner's Pocketbook** is a considered and intelligent introduction to this intriguing field, and will start the 'learner of learning' off in the right direction – on a journey that will provide immeasurable satisfactions and reward.

Tony Buzan

HOW TO USE THIS BOOK

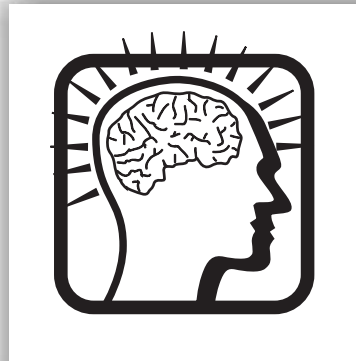
FIRST STEPS

1: List key words/headings of your existing knowledge of how to learn, write down the questions you want this book to answer and the **benefits to you** of learning how to learn.

2: Browse **quickly** through the book looking at contents, pictures, diagrams, headings, subjects. If it will **not** answer your questions or give you the benefits you seek: **DO NOT READ IT.** If it **will**:

- Ensure you are in the right physical and emotional state for learning
- Browse more slowly, taking in also the book's organisation and structure
- When ready to read in detail, choose your environment carefully, take regular breaks
- As you read, mark pages, make notes, colour in diagrams, complete the exercises
- Skip tricky text and refer back to it later
- Use what you are learning, practise and get actively involved
- When you have finished reading, summarise key points. Apply what you have learned – otherwise you have not learned

We remember: 20% of what we read, 30% of what we hear, 40% of what we see, 50% of what we say, 60% of what we do, **90% of what we read, hear, see, say and do**



BRAIN POWER

BRAIN POWER

VITAL STATISTICS



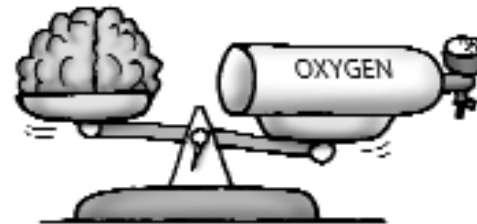
- Average adult brain weighs 3 lbs (1.4 kg) = 1 1/2 bags of sugar



- The brain is 70% water



- The brain uses 25% of the body's oxygen and blood supply, but is only 3% of bodyweight



The brain is a muscle. The more you use it, the stronger and bigger it gets. Researchers at the Institute of Neurology at University College London scanned the brains of 16 London cab drivers. They discovered that the part of the brain that stores the mental maps of the capital had grown in size. The longer the cabbie had been driving, the bigger the increase in brain size.

BRAIN POWER

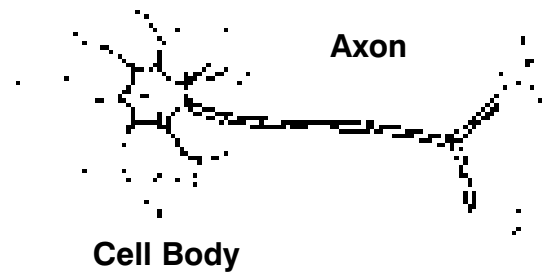
NEURONS



- The brain has 10–15 billion neurons (over twice the number of people in the world)
- It is not the number of neurons that determines intelligence but the number of **connections**
- There are at least 10 trillion (10,000,000,000,000) connections, called synapses
- A synapse is the point where the axon of one neuron connects with the dendrite of another

20,000
neurons would
fit on a pin
head

Dendrites
Nucleus



Synapse

BRAIN POWER

BRAIN WAVES



In 1929 German psychiatrist Hans Verger recorded electrical activity in the human brain. He became the *laughing stock* of the scientific world. Only three years later Edgar Adrian won a Nobel prize for demonstrating the electrical activity in the brain. We now know there are four electrical brain waves. Each oscillates at a different number of cycles per second.

Alpha Waves (Oscillating at a rate of 8–12 cycles per second)

Emitted when we are in a relaxed, meditative state, which facilitates inspiration, fast assimilation of facts and heightened memory. The ideal state for learning.

Beta Waves (Oscillating at 18–40 cycles per second)

Emitted when we are wide awake, active and figuring out complex problems.

Theta Waves (Oscillating at 4–7 cycles per second)

Emitted when we are in a state of high mental creativity – daydreaming, and the early stages of sleep, when we dream. When flashes of inspiration can occur.

Delta Waves (Oscillating at 0.5–3 cycles per second)

Emitted when we are in deep dreamless sleep or unconscious.

8

More than one brain wave can be emitted at any one time. One type will dominate, which describes the *brain state* you are in.

BRAIN POWER

THREE BRAINS



Neocortex (cortex)

► Intellectual brain

Comprises 2-sided cerebrum, left and right brain
Makes up 80% of the brain's mass
Back processes visual data, the side processes auditory data and the centre strip processes data from touch
Controls intellectual processes – talking, seeing, hearing, reasoning, thinking, problem solving



Mammalian brain (limbic system)

► Emotional brain

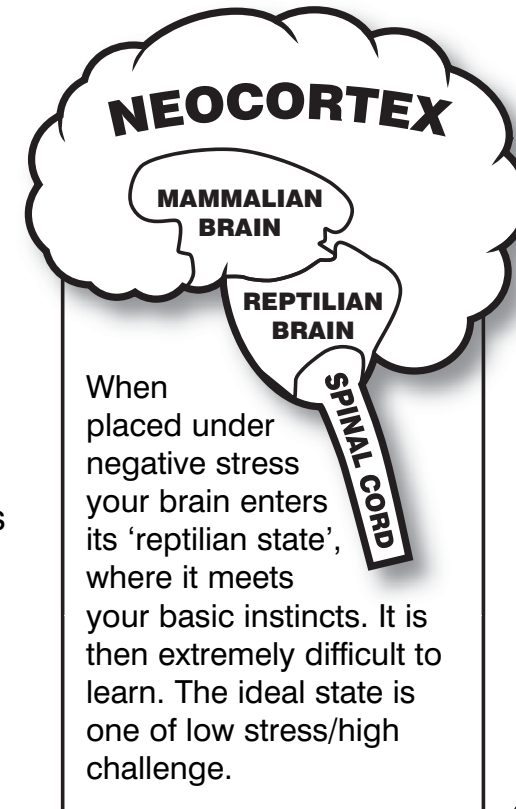
Contains hypothalamus, pituitary gland and hippo campus
Plays a vital role in long-term memory
Controls emotion, sexuality, health, immune system, sleep



Reptilian brain (brain stem)

► Instinctive brain

Stems from the spinal column
Controls basic instincts – breathing, heart rate, sense of territory, flight or fight brain



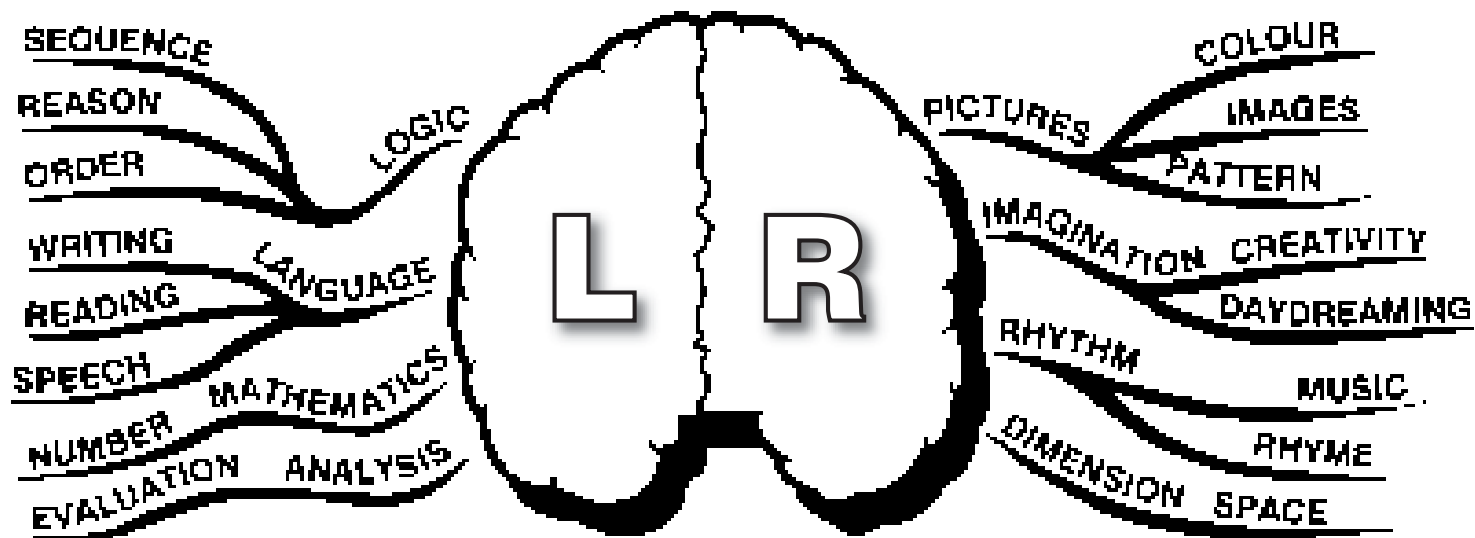
BRAIN POWER

LEFT & RIGHT BRAIN



Logical brain

a **R**istic brain



BRAIN POWER

LEFT & RIGHT BRAIN



In more recent years neuroscientists have argued that the view that the left and right hemispheres carry out different functions is too simplistic. This *lateralisation theory* is now out of date.

They favour the view that the brain carries out a wide range of functions, the precise centres of which are not yet fully mapped. Some that have been identified are:

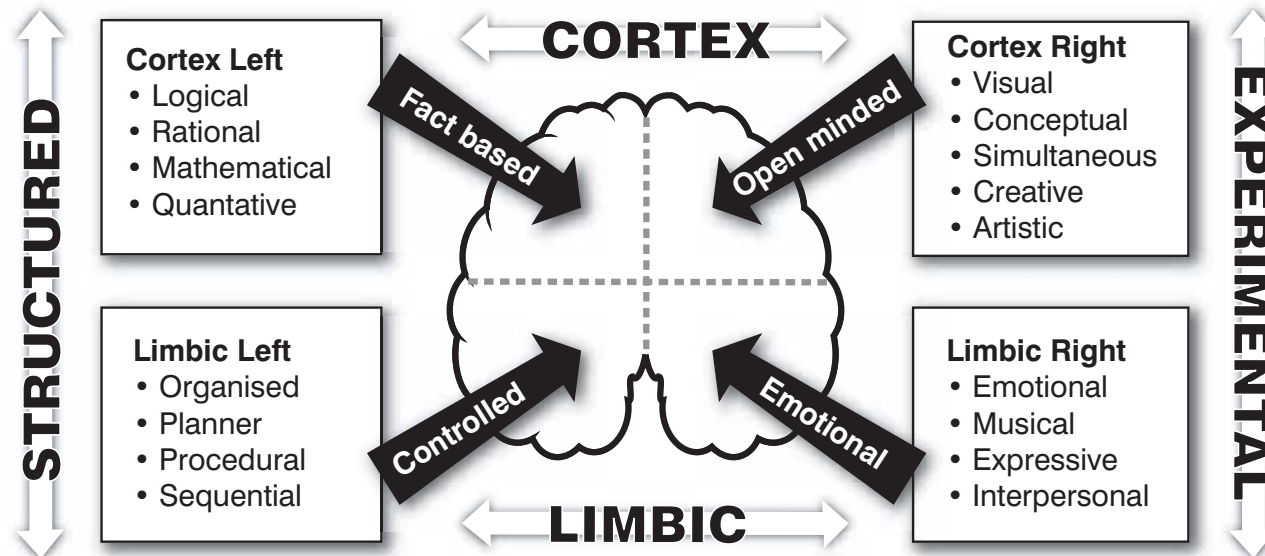
- *Wernicke's area* – manages the comprehension of the spoken language
- *Broca's area* – manages muscle control of the throat and mouth used in the spoken word
- *Somatosensory area* – receives and processes taste, plus the messages and sensations from your skin

BRAIN POWER

LEFT & RIGHT BRAIN



Further research by Ned Hermann in the 1980s identified *laterality* of the limbic system. Subsequent work has developed and proven brain quadrants, as illustrated in this simplistic diagram:



BRAIN POWER

LEFT & RIGHT BRAIN



Even though left and right brain theory has been enhanced, it is still true that combining the **L**eft and **R**ight brain will increase your

Lea**R**ning

For example:

Learning a song:

- Left brain processes the words
- Right brain processes the music

Combine both by singing

Learning with mind maps:

- Left brain processes words and order
- Right brain processes colour and images

Combine both in a mind map

BRAIN POWER

MALE & FEMALE BRAIN



Brain scans have been able to identify differences in male and female brains. The degree to which your brain conforms to the male/female pattern is determined by the level of testosterone your brain received as a foetus.

The female brain has:

- A larger Corpus Callosum, the link between the left and the right brain
- Lower level of testosterone – more likely to be compliant and co-operative
- Higher levels of dopamine – responsible for alerting the brain to information and maintaining attention. The female brain needs less stimulus to maintain attention
- Higher levels of serotonin – responsible for engaging the emotional area of the brain. The female brain is less likely to override its emotional responses
- A greater capacity for linguistic processing

Generally, females should:

- Use both hemispheres of the brain simultaneously
- Multi-task their learning, ie learn several things, or use several learning methods at once
- Use their linguistic strength in learning
- Consider a learning set

BRAIN POWER

MALE & FEMALE BRAIN



The male brain has:

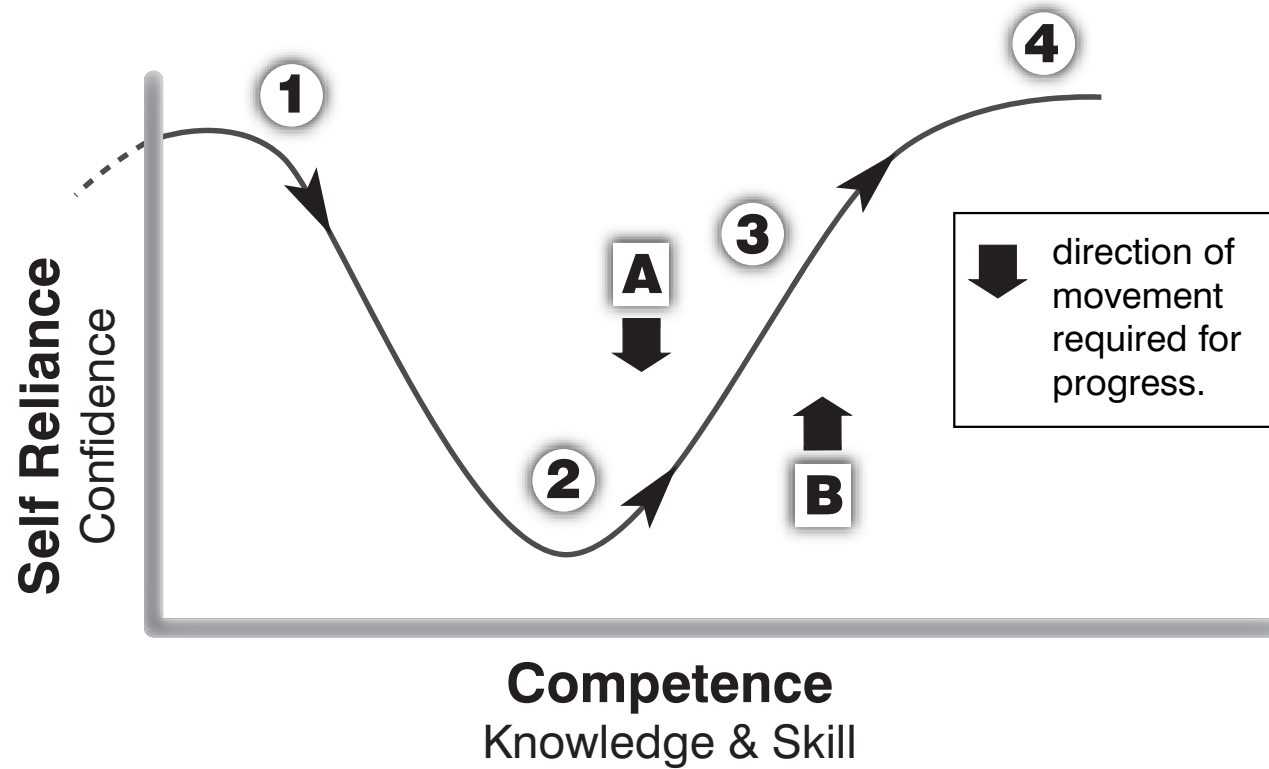
- Greater compartmentalisation – a tendency to use the left and right brain independently, focusing on one task at a time
- Higher levels of testosterone (10 x females) – more action oriented/competitive
- Lower levels of dopamine – the male brain needs greater stimulus to maintain attention
- Lower levels of serotonin – the male brain is more likely to override the emotional responses of the reptilian and mammalian brains
- A greater capacity devoted to visual-spatial processing

Generally, males should:

- Focus on a specific topic or task when learning
- Use a more structured learning style
- Use greater levels of stimulation to maintain their interest
- Set achievable targets for early and continued success
- Use their visual-spatial strength in learning

BRAIN POWER

LEARNING CURVE



BRAIN POWER

LEARNING CURVE



- 1 Unconscious incompetent** Your confidence exceeds your ability, you do not know, you are not knowledgeable/skilful
- 2 Conscious incompetent** Your confidence drops as you realise your ability is limited
- 3 Conscious competent** Your confidence increases as your ability increases, you have to concentrate on what you know/do
- 4 Unconscious competent** Your confidence and ability have peaked, you no longer have to concentrate on what you know/do; this is the start of the next learning curve. In order to learn, you must be on the curve to move forward

In different areas of your life you will be at different stages on different learning curves.

At **A** confidence exceeds ability – a drop in confidence is needed to move forward.

At **B** ability exceeds confidence – a boost in confidence is needed to move forward.

BRAIN POWER

THE FOUR STAGES OF LEARNING



Reasoning

- Focus your mind on why you are learning
- Get in a positive frame of mind



Planning

- Break the bulk of your learning down into manageable chunks
- Plan your time and environment



Committing

- Commit yourself to learn
- Put knowledge into your brain for future use



Reflecting

- Prove to yourself you know
- Get better, for future learning



REASONING & PLANNING

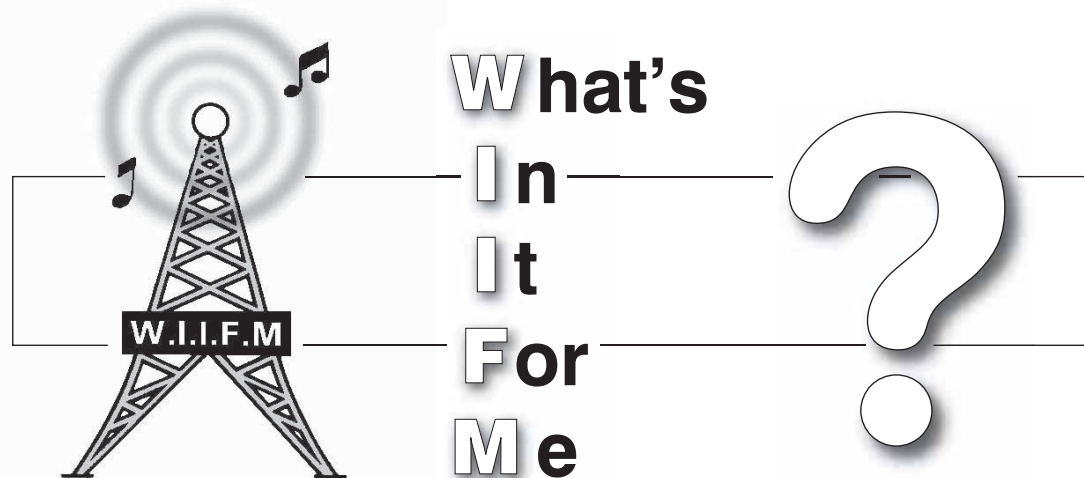
REASONING

FOCUS THE MIND



'Why am I learning?' - **If you do not know, you will not learn.**

Remember



ie: Do not make your focus passing an exam or achieving a qualification.
Focus on what that qualification gives you or will do for you.

REASONING

FOCUS THE MIND



Prepare your mind for learning:

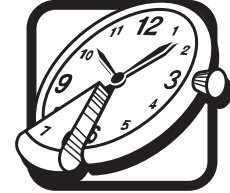
Use		affirmations	'I enjoy learning' 'I learn and remember easily' 'I am a confident learner'
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Visualise what success will feel and look like – use positive past experiences. Positive experiences are more easily remembered: make whatever you learn easier to remember.

For more information on goal setting, affirmations, visualisation and achieving success with your learning, see **The Personal Success Pocketbook**, also by Paul Hayden.

PLANNING

CREATING TIME



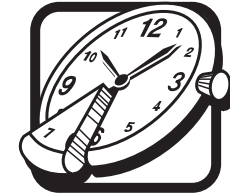
We all have the same resource in a day – 86,400 seconds. How you invest yours is important.

Create time for learning, eg:

- Rearrange activities to free up chunks of time
- Sacrifice other activities, remember WIIFM
- Switch off the TV at the end of the programme you want to watch
- Integrate study with other areas of your life
- Sacrifice lunch breaks occasionally and grab a sandwich (do not skip lunch, just the break)
- Use travel time on buses and trains
- Carry notes/books etc to study during downtime, eg: queuing

PLANNING

STRUCTURE YOUR LEARNING



Salami Method

Does not look appealing

until

you slice it up



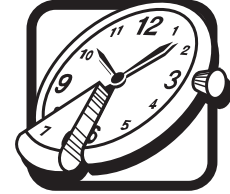
Break your learning down into manageable pieces; it makes it easier to swallow.

Backward thinking makes it easy to focus on what needs to be done daily

- Start with the end in mind
- Work back from your target/deadline date
- Spread out your slices

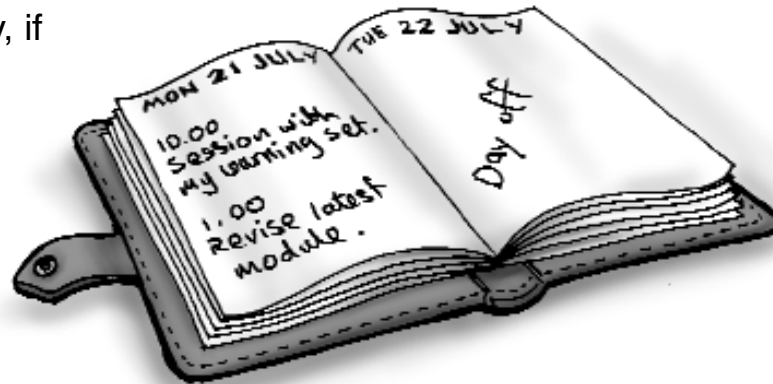
PLANNING

LEARNING PLAN



- Use your diary to plan your learning
- Make yourself a learning plan, mapping out blocks of time on particular days
- Plan days off as a reward
- Commit to learning on the days you have planned and enjoy the days off – do not overdo it

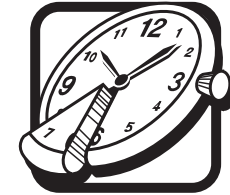
Experiment with various times of the day, if possible, to see when you learn best.



PLANNING

MIND & BODY

DIET

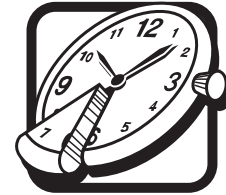


- You would not put the wrong fuel into a high performance sports car: do not put the wrong fuel in your body and mind. Too much sugar, starch, caffeine and alcohol dull the mind. (NB: tea has more caffeine than coffee. Caffeine is a diuretic and will dehydrate you)
- Drink the recommended daily allowance of water to re-hydrate your brain. Dehydration shrivels brain cells, causing headaches and impaired creativity and memory
- Recent research by Northumbria University has shown that chewing gum has a positive effect on concentration and memory and boosts the oxygen supply to the brain
- Keep a well balanced diet. As diet has become worse, more and more people are suffering from mental illness, most commonly anxiety and depression
- While you sleep your brain still burns energy, which needs replacing. Skipping breakfast results in dullness and forgetfulness for the rest of the day
- Glucose generates the 20/25 watts of electricity the brain needs to function. The brain uses two thirds of the body's glucose. Having five or six smaller meals throughout the day maintains a more even metabolic rate, aiding concentration. Don't skip breakfast or lunch

PLANNING

MIND & BODY

DIET – VITAMINS & MINERALS



Vitamin A – Aids vision; found in fats and oils

B Vitamins – Keep you mentally alert; found in wholegrains, seeds, lentils, yeast, nuts, eggs and milk products

Vitamin B3 is essential for proper brain functioning

Vitamin B5 and **B6** are particularly important for memory; also found in fish and chicken

Vitamin B12 is essential for the production of red blood cells, which carry oxygen.

Found in animal products such as meat, milk and eggs.

Vegetarians may need to take extra as it is scarce in their diet

Vitamin C – Anti-oxidant; protects other vitamins from destruction. Also contains acetylcholine, a neural transmitter essential for short and long-term memory function; found in fresh fruit and vegetables

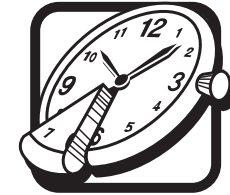
Vitamin E – Increases cell oxygenisation. Protects the brain against oxidised fats that slow the brain metabolism; found in wholegrains, whole wheat and sunflower seed oil

Iron – Used to transport oxygen in the blood. Remember, your brain uses 25% of your body's oxygen. Also assists in the promotion of cell and tissue growth, including in the brain; found in meat and fresh vegetables

PLANNING

MIND & BODY

RELAXATION



Energy for learning is released through relaxation. Before learning, bring about a calm and positive mood. Try the following:

With your eyes closed, sit and:

- Listen to relaxing music
- Tense and relax each muscle in turn, starting with your feet and working up your body
- Imagine walking down a flight of stairs; after each step exhale and say, 'I am even more relaxed now'

These exercises have a double effect as they also stimulate both sides of the brain in preparation for learning.

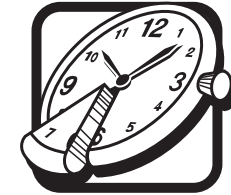
Sleep

Research has shown that sleep deprivation reduces your ability to learn and retain information. It also shows that on average we get one to two hours less sleep than we need. Sleep also enables your brain to process what you have learned and make sense of it.

PLANNING

MIND & BODY

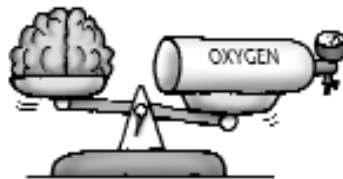
EXERCISE & MOVEMENT



Movement stimulates the flow of blood and oxygen to your brain. It gives the body more sensory cues to trigger and aid recall. Motion determines emotion. The word 'emotion' comes from the Latin *movere* meaning move.

Your body language communicates the way you feel. The reverse is also true. Your body language can be used to influence the way you feel. When you are learning, sit appropriately and get up and move about regularly.

- Exercise regularly to clear the arteries and oxygenate the blood
- Remember the brain uses 25% of body's oxygen and blood supply
- Controlled rhythmic breathing results in greater brain oxidation
- Use what exercise works best for you, but regular **rest** is also necessary for the brain to function fully



*'MENS SANA IN
CORPORE SANO'*

A healthy mind in
a healthy body

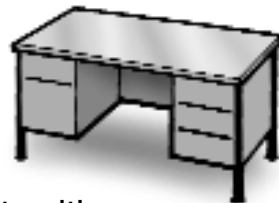
PLANNING

ENVIRONMENT



Workspace

- Should be inviting and encourage study
- Remove all distractions
- Ensure everything you need is to hand eg books, pens and paper
- Display posters and mind maps to stimulate your mind



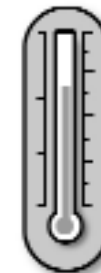
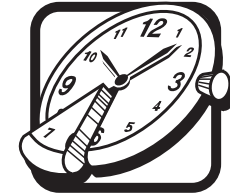
Table/Desk

- Navel height, with as much space as necessary
- Do not cramp yourself



Chair

- Comfortable and straight backed
- High enough so your feet are flat on the floor and thighs parallel to the ground
- Good posture increases supply of blood to the brain

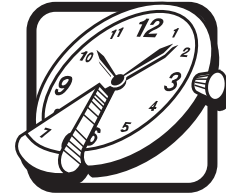


Temperature

- Not too warm, otherwise you relax

PLANNING

ENVIRONMENT



Light

- Natural light is best – sit by a window if you can
- Daylight affects your melatonin and hormone levels. These influence the release of neurotransmitters, essential to memory and alertness. The more daylight you get, the more alert you will be
- Standard bulb is better than fluorescent light



Music

- See page 93



Air

- To function well the brain needs oxygen – as clean and fresh as possible; open the nearest window
- At breaks go outside if possible
- Plants oxygenate a room, filter out harmful gases from the air and create an air of calm



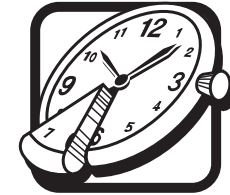
Toys

- Try squeezing stress balls, playing with slinkies, fidgeting with bendy toys, etc.
- Touch stimulates the nerve endings near the skin's surface. These nerve endings send messages to the brain, keeping your brain alert

PLANNING

ENVIRONMENT

FRAGRANCE & LEARNING



Dr Alan Hirsch of the Smell and Taste Research Foundation in Chicago noted an increase from 14% to 54% for students in a calculus exam, when they were exposed to a floral aroma. The sense of smell has unfiltered, quicker access to the brain than any of the other senses. It can raise attention and awareness levels dramatically. Think of freshly baked bread or fresh coffee.

Here is a list of the characteristics of various aromas:

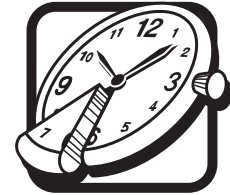
Grapefruit	Radiating, cheerful, liberating
Lemon	Stimulating, clarifying, concentration-building
Narcissus	Empowering, visionary, creative
Peppermint	Clarifying, awakening, stimulating, refreshing
Sandalwood	Enlightening, balancing, connecting
Thyme	Empowering, assisting, invigorating

(See further: The Fragrant Mind by aromatherapist Valerie Ann Worwood.)

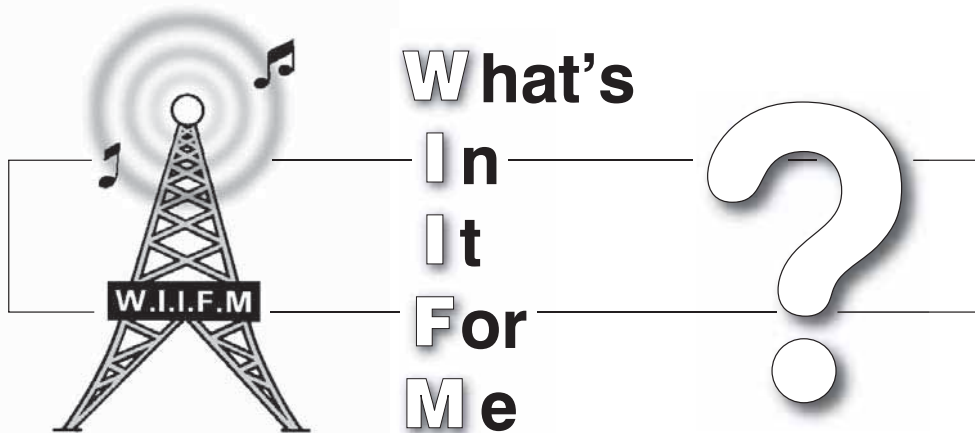
Consider using scented gel pens or markers to introduce colour and scent to your learning.

PLANNING

RE-FOCUS



When you are ready to start learning, re-focus.



'Why am I learning?'



COMMITTING

COMMITTING

STARTING FROM SCRATCH? BEFORE BEGINNING TO LEARN



- Write down key headings/words of what you already know, or use a mind map to provide the links for the new information.
- In the unlikely event you know absolutely nothing:
 - Read the appropriate section of an encyclopedia, or
 - Read a book aimed at children; these are especially good as they are simple and have plenty of pictures to stimulate the right brain too
- Use brain gym exercises (see next page)



COMMITTING

BRAIN GYM



These are warm up activities to prepare the brain for learning and/or as energisers during learning. If the brain is warmed up, the information being learned will be retained more effectively.

A warm up can include a recap on existing knowledge and activities that stimulate both hemispheres of the brain. The ideal study time for an adult is 20–30 minutes but taking a break every half hour may be disruptive. Instead use a brain gym exercise to re-energise yourself.

Brain gym exercises also stimulate alpha brainwaves. This induces the ideal state for learning.

Brain gym exercises

- Logic problems
- Memory games
- Brain teasers
- Rebus
- Pat your tummy and rub your head
- Juggling
- Quick puzzles
- Recite the alphabet backwards

COMMITTING

THE SIX W'S

BEFORE BEGINNING TO LEARN



Create an interest in the material to be learned. Remember – interest creates motivation to learn and aids retention.

Ask questions that you want answered by your study.

- | | |
|----------------|--|
| W ho? | Who discovered this? Who are the key characters? |
| W hat? | What are the counter arguments? What are the key facts? |
| W hen? | When was this discovered? When did this happen? |
| W here? | Where did this happen? Where was he/she born? |
| W hy? | Why did this happen? Why should I believe this? |
| HoW ? | How will this work? How does this relate to my existing knowledge? |

If you are not asking questions you are not learning.

COMMITTING

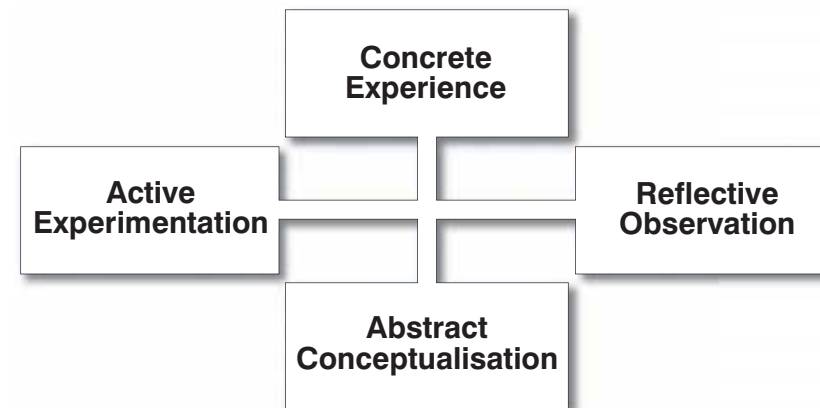
KOLB'S CIRCULAR LEARNING THEORY



David Kolb developed an experiential learning theory:



Kolb's theory was built around two dimensions:



His theory is that effective learning requires the use of all four abilities. Few people are equally balanced, most tend to develop a strength, or preference. Using only your preferred style can put you at a serious disadvantage.

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES



ACTIVISTS

Peter Honey and Alan Mumford built on Kolb's work, developing four learning styles. The following will help you identify your preferred style, get the most from it and show you how to develop a more rounded style:

Activists – Preference for active experimentation (Philosophy: 'I'll try anything once')

Activists:

- Involve themselves fully and without bias in new experiences
- Tackle things with a *have a go* mentality
- Enjoy and live in the here and now
- Don't need detail
- Are open minded/optimistic, love anything new
- Act first and consider the consequences later
- Can take risks for the sake of it
- Thrive on new challenges but get bored quickly
- Are extrovert, enjoy group activities and performing to a crowd
- May become glory seekers
- Are not good at finishing
- May find training sessions too slow

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES



ACTIVISTS

You learn **best** from activities that:

- Have elements of *have a go* and offer short *here and now* exercises and games
- Provide new experiences and opportunities to learn from
- Allow you to get involved with others, eg solving problems as a team
- Have assorted activities to tackle, that change quickly, are challenging and provide opportunities for glory
- Present you with the opportunity to be in the limelight, eg as a spokesperson for your group, facilitate a discussion

You learn **least** from activities that:

- Require you to be passive, eg reading, listening to a presentation
- Involve solitary work
- Repeat the same routine and predictable tasks including excessive practising
- Demand attention to detail, a thorough job and analysis of data
- Give you little room for movement because they are so precise

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

REFLECTORS



Reflectors – Preference for reflective observation (Philosophy – ‘Let me think about this’)

Reflectors:

- Prefer to stand back to ponder experiences and consider them from many angles
- Prefer to collect information from different sources and consider alternatives before coming to a conclusion
- Are happy to deal with detail
- Are cautious and thoughtful
- Are holistic – like to look at all sides of the picture
- Are unable to make quick decisions
- Take a back seat in meetings and debates
- Enjoy observing other people in action
- Listen to others' points of view before making their own points
- Enjoy discussing and putting things into perspective
- Adopt a low profile and have slightly detached, aloof, calm characters
- Are happy to work behind the scenes
- Prepare well
- May find training sessions too fast

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

REFLECTORS



*You learn **best** from activities that:*

- Encourage you to watch, listen and think, eg observing a roleplay or watching a video
- Allow you time to think and prepare fully, before you decide/act
- Use activities such as research, observation, analysis of data and regular reviews
- Get you to produce considered and measured analyses and reports
- Allow you to exchange thought out views in a low-risk environment
- Offer opportunities for feedback

*You learn **least** from activities that:*

- Force you into the limelight, eg roleplays, particularly if they allow no planning time
- Get you doing something without warning, eg to give an instant reaction or idea
- Have restrictive timeframes, fast pace or rush you from one exercise to the next
- Have poor communication and information flow, requiring you to make decisions with insufficient information, eg a badly briefed exercise on a training course
- Give no opportunity for reviews or feedback

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

THEORISTS



Theorists – Preference for abstract conceptualisation (Philosophy – ‘What’s the relationship?’)

Theorists:

- Assimilate and adapt observations and abstract ideas into complex yet logically sound and coherent theories
- Are reluctant to test theories in practice
- Think problems through in a step-by-step rational, logical way but can over-complicate
- Only produce prepared ideas – low risk takers
- Are perfectionists who won't rest until things fit into a rational scheme
- Like to investigate and integrate
- Comfortable with principles, theories, models and systems
- Prefer to work alone and take their time
- Prefer rational objectivity to subjective ambiguity
- Are uncomfortable with lateral thinking and anything flippant
- Have the patience for complexities
- May have an ivory tower image
- Ask questions probing basic assumptions

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

THEORISTS



*You learn **best** from activities that:*

- Use, or are part of, a model, theory or philosophy
- Give you time to explore thoroughly and logically the links, relationships and associations
- Allow you to investigate underpinning fundamentals, logic or theory
- Stretch you intellectually, eg analysing, debating or participating in complex situations to test your understanding
- Provide clear guidelines, structure and purpose
- Give you valuable concepts, even if they are not relevant
- Involve well argued, watertight debate

*You learn **least** from activities that:*

- Are vague or uncertain and lack structure, context or reason
- Put emphasis on feelings and emotions, eg some teambuilding events
- Are results and action oriented, especially against tight deadlines
- Skim the surface, offering a high level view, without depth or substance
- Have contradictory methods/principles

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

PRAGMATISTS



Pragmatists – Preference for concrete experience (Philosophy – ‘What would happen if...’)

Pragmatists:

- Are keen to try out new ideas, theories and techniques to see if they work in practice
- Have a commonsense, practical approach to making things work
- Like trial and error
- Positively search out new ideas and take the first opportunity to experiment with applications
- Need some detail – only if they think it relevant and useful
- Impatient with reflecting, open-ended discussions and theory
- Prefer to get straight to the point in a discussion, they say what they think
- Like to get things done efficiently
- Will break rules if there is a better way to do things
- Are problem solvers – produce spontaneous ideas
- May make hasty decisions
- See problems and opportunities as a challenge

COMMITTING

HONEY AND MUMFORD'S LEARNING STYLES

PRAGMATISTS



*You learn **best** from activities that:*

- Give you a model to copy, eg a demonstration before you have a go, examples and stories that illustrate the point
- Allow you to practise with feedback from someone with proven credibility
- Have a clear and practical link between the learning and the application
- Present an immediate opportunity to implement the learning
- Offer clear practical benefits, eg learning how to learn

*You learn **least** from activities that:*

- Bear no resemblance to reality, eg some training games
- Do not offer you any benefits, eg saving, gain, guarantees
- Offer no opportunity for you to practise
- Are not able to be implemented quickly or that have barriers to implementation, eg management, red tape, procedures
- Have a high cost of failure

COMMITTING

LEARNING STYLES

THREE KINDS OF MEMORY



Your brain receives information and remembers it through the five **senses**:

- Sight (visual memory)
- Touch (kinaesthetic memory)
- Sound (auditory memory)
- Taste (gustatory memory)
- Smell (olfactory memory)

Learning studies have identified three primary kinds of memory:

Visual, Auditory and **Kinaesthetic**



Visual learners find it easier to take in new information through pictures, diagrams, charts, films, etc



Auditory learners find it easier to take in new information through the spoken word



Kinaesthetic learners find it easier to take in new information through copying demonstrations and getting physically involved

Controlled by the Neocortex, see page 9.

COMMITTING

LEARNING STYLES

THREE KINDS OF MEMORY



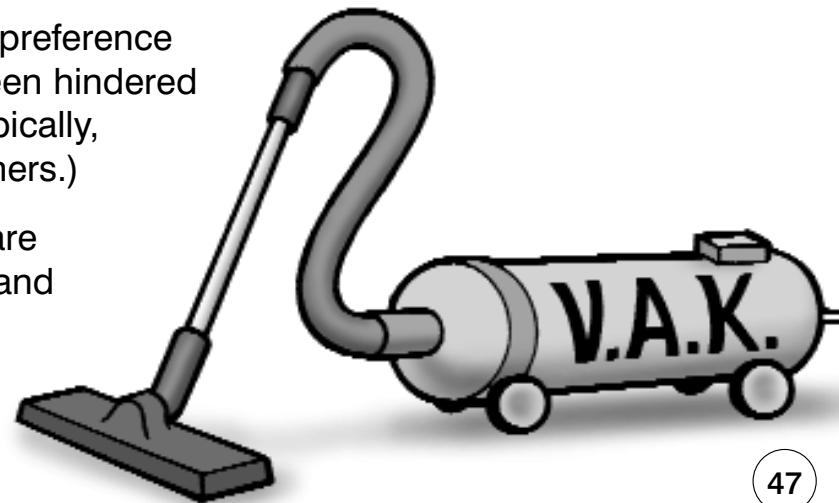
The best learning takes place using all three memories, eg:

- If you are reading:
- Visualise the key messages
 - Read aloud or hear the words internally
 - Get physically involved – underline, highlight, mind map, etc

Use all three styles to **V.A.K.** up knowledge.

Learners dominate in one style and have a preference for another. Previous learning may have been hindered if it did not cater for your learning style. (Typically, schools are not geared to kinaesthetic learners.)

Research has proven that 29% of learners are primarily visual, 34% are primarily auditory and 37% are primarily kinaesthetic. Assess your style on the following pages and use your preferred style(s) to learn, using the following tips.



COMMITTING

LEARNING STYLES

VISUAL LEARNERS



- Use phrases such as 'I **see** what you mean', 'That **looks** right'
- When relaxing, prefer to watch a film or video, go to the theatre or read
- Prefer to talk to people face to face
- Are fast talkers, dislike listening to others
- Forget names, remember faces
- If lost or need directions, prefer a map
- When inactive, tend to doodle or watch someone/something
- When angry, are silent and seethe
- Reward people with a note, letter or card
- Are well dressed, tidy and organised

LEARN BEST BY:

- Writing down key facts or, better still, making mind maps
- Visualising what they are learning by creating pictures/diagrams
- Using time lines, for remembering dates
- Creating their own strong visual links
- Using pictures, diagrams, charts, film, video, graphics, etc

COMMITTING

LEARNING STYLES

AUDITORY LEARNERS



- Use phrases such as 'That **sounds** right', 'I **hear** what you are saying'
- When relaxing, listen to music or radio
- Prefer to talk to people on the phone
- Enjoy listening to others, but impatient to talk; talk in a rhythmic voice
- Forget faces, remember names
- If lost or need directions, prefer to be told
- When inactive, tend to talk to themselves or others
- When angry, express themselves in outbursts
- Reward people with oral praise
- Do not like reading books or instruction manuals

LEARN BEST BY:

- Hearing a seminar, presentation or explanation
- Reading aloud to themselves with emotion or accent
- Making a tape of key points to listen to in the car, while ironing, etc
- Verbally summarising in their own words
- Explaining the subject to someone else
- Use their own internal voice to verbalise what they are learning

COMMITTING

LEARNING STYLES

KINAESTHETIC LEARNERS



- Use phrases such as 'I found it easy to **handle**', 'That **touched** a nerve'
- When relaxing, prefer to play games and sport
- Prefer to talk to people while doing something else
- Slow talkers, use gestures and expressions
- Shake hands with people they meet
- If lost or need directions, prefer to be shown the way
- When inactive, cannot sit still for long
- When angry, clench their fists, grit their teeth and storm off
- Reward people with a pat on the back

LEARN BEST BY:

- Copying demonstrations
- Making models
- Recording information as they hear it, preferably in a mind map
- Walking around while they read
- Underlining/highlighting new information/key points
- Putting key points on to index cards and sorting them into order
- Getting physically and actively involved in their learning

COMMITTING

EIGHT OR MORE INTELLIGENCES

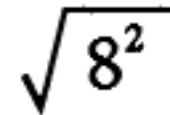


The concept of multiple intelligences was first discovered by Professor Howard Gardner of Harvard University in 1983. His ground-breaking book, *Frames of Mind* redefined beliefs on human intelligence. Gardner's research originally introduced seven intelligences. He was able to accurately pinpoint the area of the brain that correlated to each of these intelligences. In 1999 he qualified and added an eighth: Naturalistic Intelligence.

Linguistic Intelligence – used for reading, writing and speech



Logical-Mathematical Intelligence – used for Maths, logic and systems



Visual-Spatial Intelligence – used for visualisation and art



Musical Intelligence – used for rhythm, music and lyrics



COMMITTING

EIGHT OR MORE INTELLIGENCES



Bodily-Kinaesthetic Intelligence – used for touch and reflex



Interpersonal Intelligence – used for communicating with others



Intrapersonal Intelligence – used for self-discovery and self-analysis



Naturalistic Intelligence – used for making sense of the natural world



In addition, Gardner and his colleagues also suggest the following intelligences may exist:

*Spiritual intelligence *Moral intelligence *Digital intelligence, and most convincingly, they argue there is *Existential intelligence – a concern with ultimate issues.

COMMITTING

EIGHT OR MORE INTELLIGENCES



Everyone possesses all the intelligences to some extent.

The most powerful learning combines all of them.

All intelligences are of value, you need to identify where your strengths lie and use those intelligences.

Whether these are intelligences or abilities, as Professor Howard Gardner puts it, 'Multiple intelligence education is multiple chance learning'.

Assess your strengths ☐ on the following pages and use them by combining the different learning techniques.

COMMITTING

EIGHT OR MORE INTELLIGENCES

LINGUISTIC INTELLIGENCE



- Used for reading, writing and speech

Characteristics:

- ☐ Extensive vocabulary, good at spelling
- ☐ Good verbal and/or written communication
- ☐ Expressive, fluent talker, gives clear explanations
- ☐ Good listener
- ☐ Reasoning ability
- ☐ Methodical

Likes:

- ☐ Reading books/magazines/newspapers
- ☐ Word games/crosswords/scrabble
- ☐ Word play/puns/tongue twisters
- ☐ Theatre, radio
- ☐ Poetry
- ☐ Debate
- ☐ Writing letters

English and History were favourite subjects at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

LINGUISTIC INTELLIGENCE



Learning techniques:

- Learn from books, tapes, lectures, presentations, seminars, etc
- Write down questions you want answered before starting any learning
- Read out loud
- After reading a piece of text, summarise in your own words out loud and write it down
- Always put things into your own words
- Brainstorm to organise thoughts into order and/or key points
- Write key points on cards and sort into order
- Make up crosswords and puzzles to solve (why not do this with your learning set?)
- Debate and discuss issues (preferably with your learning set)
- Present what you have learned orally or in writing to someone else (your learning set?)

COMMITTING

EIGHT OR MORE INTELLIGENCES

LOGICAL-MATHEMATICAL INTELLIGENCE



- Used for Maths, logic and systems

Characteristics:

- ☐ Good at budgeting
- ☐ Logical thought, explanation and action
- ☐ Organised, organises tasks into sequence
- ☐ Plans time and journeys effectively
- ☐ Reasoning ability
- ☐ Seeks patterns and relationships
- ☐ Precise

Likes:

- ☐ Calculations, eg dart scores, gambling odds, etc
- ☐ Solving puzzles/brain teasers requiring logical thinking
- ☐ Putting together a detailed itinerary for holidays and business trips
- ☐ Abstract thought
- ☐ Experimenting
- ☐ Scientific advances
- ☐ Computers

Maths and Science were favourite subjects at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

LOGICAL-MATHEMATICAL INTELLIGENCE



Learning techniques:

- List key points in order and number them
- Use a flow chart to express information/knowledge in easy to follow steps
- Use mind maps
- Use computers, eg spreadsheets
- Experiment with the knowledge
- Use timelines for remembering dates and events
- Analyse and interpret data
- Use your reasoning and deductive skills
- Create and solve problems (this can be done with your learning set)
- Play mathematical games (this can be done with your learning set)

COMMITTING

EIGHT OR MORE INTELLIGENCES

VISUAL-SPATIAL INTELLIGENCE



- Used for visualisation and art

Characteristics:

- ☐ Thinks and remembers in pictures
- ☐ Good sense of imaging/use of mind's eye
- ☐ Sense of colour, good at art/drawing
- ☐ Uses maps, charts and diagrams easily
- ☐ Sense of direction
- ☐ Good at driving/parking
- ☐ Well dressed

Likes:

- ☐ Film and video
- ☐ Posters/pictures
- ☐ Drawing, painting, sculpting
- ☐ Doodling
- ☐ Colour
- ☐ Clothes, dressmaking
- ☐ Self assembly furniture/jigsaw puzzles and mazes
- ☐ Navigating
- ☐ Photography

Art was a favourite subject at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

VISUAL-SPATIAL INTELLIGENCE



Learning techniques:

- Learn from film, video, slides, etc
- Use symbols, doodles, diagrams or, better still, mind map
- Design and produce a poster of the key facts and pin it up
- Highlight key points with different colours
- When you read, visualise events in your mind's eye; do not focus on the words
- Use heavily illustrated material to learn from
- Use visualisation to create a mental TV documentary with strong visual images
- Study in different places/areas of a room to gain a different perspective
- Convert information into diagrams or cartoons

COMMITTING

EIGHT OR MORE INTELLIGENCES

MUSICAL INTELLIGENCE



- Used for rhythm, music and lyrics

Characteristics:

- ☐ Sensitive to pitch, rhythm and timbre
- ☐ Sensitive to emotion of music
- ☐ Changes mood with music
- ☐ Good at clapping in time to music
- ☐ Moves in time to music
- ☐ Remembers and repeats slogans and lyrics easily
- ☐ Theme music or jingles often pop into mind
- ☐ Good at selecting background music
- ☐ May be deeply spiritual

Likes:

- ☐ Radio
- ☐ Concerts
- ☐ CD collection
- ☐ Making music – plays an instrument
- ☐ Singing – in a choir, group or alone
- ☐ Writing songs and/or music
- ☐ 'Working out' or relaxing to music

Music was a favourite subject at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

MUSICAL INTELLIGENCE

Learning techniques:

- Use music to relax before learning
- Study to music that represents what you are learning
- Study to baroque music (see page 93)
- Read rhythmically (use a metronome)
- Write a song, jingle, rap, poem, rhyme, etc, to summarise key points



COMMITTING

EIGHT OR MORE INTELLIGENCES

BODILY-KINAESTHETIC INTELLIGENCE



- Used for touch and reflex

Characteristics:

- ☐ Never sits still
- ☐ Mechanically minded
- ☐ Likes to touch
- ☐ Solves problems physically hands on
- ☐ Good with their hands
- ☐ Controlled reflexes
- ☐ Control of body
- ☐ Control of objects
- ☐ Good timing

Likes:

- ☐ Sport/games
- ☐ Rough and tumble play/thrill rides at the fair
- ☐ Acting/drama
- ☐ Dancing
- ☐ Cooking/baking
- ☐ Handicrafts
- ☐ D.I.Y.
- ☐ Car maintenance

PE and handicraft lessons were favourite subjects at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

BODILY-KINAESTHETIC INTELLIGENCE



Learning techniques:

- Tackle learning *hands on*
- Learn from what you do
- Use roleplay/drama to act out what you are learning
- Get involved in the subject physically/use field trips
- Take action – write down key points or, better still, mind map
- Make models
- Write key points on to index cards and sort them into order/groups and/or pin them up in your study area
- Move about while you are learning
- Change activity often and take frequent breaks
- Mentally review your learning while jogging/swimming/walking, etc

COMMITTING

EIGHT OR MORE INTELLIGENCES

INTERPERSONAL INTELLIGENCE



- Used for communicating with others

Characteristics:

- ☐ Relates to and mixes well with others
- ☐ Puts people at ease
- ☐ Has numerous friends
- ☐ Sympathetic to others' feelings
- ☐ Mediates between people in dispute
- ☐ Good communicator
- ☐ Good at negotiating
- ☐ Co-operative

Likes:

- ☐ Being with people
- ☐ Parties and social events
- ☐ Community activities
- ☐ Clubs
- ☐ Committee work
- ☐ Group activities/team tasks/games
- ☐ Managing/supervising
- ☐ Teaching/training
- ☐ Parenting

COMMITTING

EIGHT OR MORE INTELLIGENCES

INTERPERSONAL INTELLIGENCE

Learning techniques:

- Learn from others (form a learning set)
- Work in teams and learn together
- Talk to others to get and share answers
- Compare notes after a study session
- Make use of networking and mentoring
- Teach others
- Socialise during breaks
- Throw a party to celebrate/reward your success



COMMITTING

EIGHT OR MORE INTELLIGENCES

INTRAPERSONAL INTELLIGENCE



- Used for self-discovery and self-analysis

Characteristics:

- ☐ Understands own feelings and behaviour
- ☐ Self-intuitive, knows own strengths
- ☐ Private
- ☐ Independent
- ☐ Wants to be different from the crowd
- ☐ Keeps a diary/journal
- ☐ Plans time effectively
- ☐ Is self-motivated
- ☐ Sets and achieves goals

Likes:

- ☐ Peace and quiet
- ☐ Daydreaming
- ☐ Reflecting/reminiscing
- ☐ Independence
- ☐ Achieving goals
- ☐ Own company
- ☐ Solitary pursuits
- ☐ Holidaying in a country cabin (not a busy hotel)
- ☐ Working for self (or has seriously contemplated it)

COMMITTING

EIGHT OR MORE INTELLIGENCES

INTRAPERSONAL INTELLIGENCE



Learning techniques:

- Use personal affirmations (see page 21)
- Set and achieve goals/targets with your learning
- Create personal interest, why the subject matters to you
- Get interested, involved and motivated with the arguments and main characters
- Take control of your learning
- Carry out independent study
- Seek out background information, especially the human interest angle
- Listen to your intuition
- Reflect, write or discuss what you experienced and how you felt
- Reflect on how the information fits in with your existing knowledge and experiences

COMMITTING

EIGHT OR MORE INTELLIGENCES

NATURALISTIC INTELLIGENCE



Characteristics:

- ☐ Attuned to the natural world
- ☐ Good at identifying and classifying
- ☐ Can remember habitats of plants and animals
- ☐ Comfortable in different types of environments
- ☐ Troubled by pollution
- ☐ Keen sensory skills – sight, sound, smell, touch, taste

Likes:

- ☐ The outdoors
- ☐ Camping, hiking or gardening, etc
- ☐ Being close to wild animals and plants
- ☐ Observing and identifying wildlife
- ☐ Exploring and expeditions
- ☐ Taking care of living things
- ☐ Collections, scrapbooks or logs
- ☐ TV programmes, books and videos about the natural world
- ☐ Astronomy, evolution and the environment

Science was a favourite subject at school.

COMMITTING

EIGHT OR MORE INTELLIGENCES

NATURALISTIC INTELLIGENCE



Learning techniques:

- Learn outdoors, eg on field trips and practical work
- Use tools for investigating, eg magnifying glass, microscope and binoculars
- Collect and organise data from observations
- Sort things and place them in hierarchies
- Look for patterns, similarities, make connections
- Classify, discriminate or look for differences
- Stimulate your senses, eg use accelerated learning toys to stimulate your sense of touch as you learn, use scented pens and music
- Have plants and fish where you learn
- Get plenty of fresh air and even learn outdoors

COMMITTING

POWER READING

PREPARATION



- Start with introductory text such as an encyclopedia or children's book
- Find the three best books in your area of interest (ask for referrals)
- List key headings of existing knowledge or mind map it
- Define your questions – the six W's (see page 36) so you read with a sense of interest
- Browse through each book casually and rapidly to identify:
 - organisation and structure
 - appendices
 - contents, etc
 - diagrams/pictures
 - preface
- Be selective, do not go further if it does not add to your existing knowledge
- Organise your reading into chunks (salami method)
- When you take breaks:
 - cup and uncup your hands over your eyes
 - focus on distant objects
 - do not rub your eyes

COMMITTING

POWER READING

HOW TO READ



- Scan read for ideas and principles, not individual words
- Read the text; concentrate on beginning and end of paragraphs and chapters, or wherever else you identify key ideas are communicated
- Mark pages, highlight, underline, make notes, use codes as you go
- Skip over difficult text, return to it later when you have the full picture
- Read difficult text aloud
- Make key notes or mind map at the end of each chapter
- Review your notes/mind map
- Summarise the whole book in a few key notes/master mind map

TIPS

- Do not be afraid to disregard the book at any stage
- Never feel obliged to read a book cover to cover: only read what you want
- Be flexible; adapt your reading to the style of the material

Remember to celebrate your success and reward yourself.



COMMITTING

POWER READING

Try this simple test – answer all questions:

- 1 What is your favourite book?
- 2 Which spelling is correct: feasability or feasibility?
- 3 What was your favourite subject at school?
- 4 Work out what X is: $2 \times X + 3 = 15$
- 5 What did you have for breakfast?
- 6 Write your name backwards
- 7 What is your favourite film?
- 8 Repeat question 5
- 9 Ignore all the questions. If you answered, then see 10 below, if not turn the page
- 10 Re-read **Power Reading**: time spent now may save time in the future



COMMITTING

LEARNING SET

BENEFITS



Try getting together with a group to learn. The ideal number is two to six people.

- Makes learning enjoyable
- You give each other mutual support
- Saves time, as you will not have to research the whole subject
- Mixing learning styles ensures better learning
- Gain different opinions and viewpoints
- Raises the quality of your work
- You are committed if you are participating and contributing
- You will also learn how to work as a group, communicate and present your ideas
- Learning is improved – two heads are better than one

COMMITTING LEARNING SET



Reasoning

- Establish a working contract for learning together
- Define the common objective and goal

Planning

- Break the learning into sections and assign to the set members
- Complete a learning plan

Committing

- Support each other, share opinions, experiences, knowledge and fears
- Use individuals' strengths, especially learning styles

Reflecting

- Present your section to the set, proves if you know the subject
- Review how the set worked, to improve for the future
- Keep the set together for future learning projects
- Compete with other learning sets

COMMITTING

ON COURSE TECHNIQUES



Preparation

- Complete any required pre-course study
- If there is none, gain a basic understanding of the subject

Questions

- Prepare questions, know what you want to achieve
- Use the six W's

Seating

- Front and centre; you are more involved and can more easily focus on the trainer
- From this position visual aids are easier to see, speech is easier to hear
- Watch your posture; sit upright and attentive

Trainer

- Concentrate on the message not the messenger: ignore any irritating mannerisms
- If you have done your preparation and prepared questions this will be easier

Listen

- Imagine the trainer is talking to you personally
- Do not read notes or ask colleagues to explain during a training session
- If you have missed a point ask the trainer

COMMITTING

ON COURSE TECHNIQUES



Notes

- Mind map the key points; this will free up your time to listen

Learning

- Use your own personal learning style
- If this is not possible during a learning session, you have some post-course work to do

Breaks

- If you can, go outside for fresh air and a walk. Remember how daylight affects your alertness (page 30)
- Drink water, not tea or coffee (the brain is 70% water)
- Do not discuss what you are learning (let your subconscious process it)
- Have a light lunch and give your body time to digest it before re-entering the training room

Colleagues

- At the end of day or course share your learning: you will be surprised at what you have missed and gain a different perspective

Review/ Consolidate

- At the end of the day or course review your learning
- Follow up action points and keep your learning current and fresh

COMMITTING

LEARNING TRANSFER GRID



Fortune magazine undertook research with the 500 highest performing companies in the USA, asking about training. From the answers they created this grid, which reflects which areas contribute most significantly to the success of those companies. The research identified the most important elements of training in terms of enabling delegates to change their behaviour permanently, incorporating new skills into their working practices.

The best course in the world will be useless unless some part of the content is put into practice afterwards. Conversely, act on a single new skill/piece of information learned from a poor course, and you will have developed and become more effective.

An ongoing programme of post-course activity is vital to ensure that you gain optimum benefit from your attendance.

Input from	Before training	During training	After training
Supervisor	3	8	2
Delegate	6	4	1
Trainer	7	5	9

- 1 = The area where the greatest emphasis needs to be, to ensure positive benefits from training.
9 = The area where the least emphasis needs to be, to ensure positive benefits from training.

COMMITTING

MEMORY

WHAT IS MEMORY?

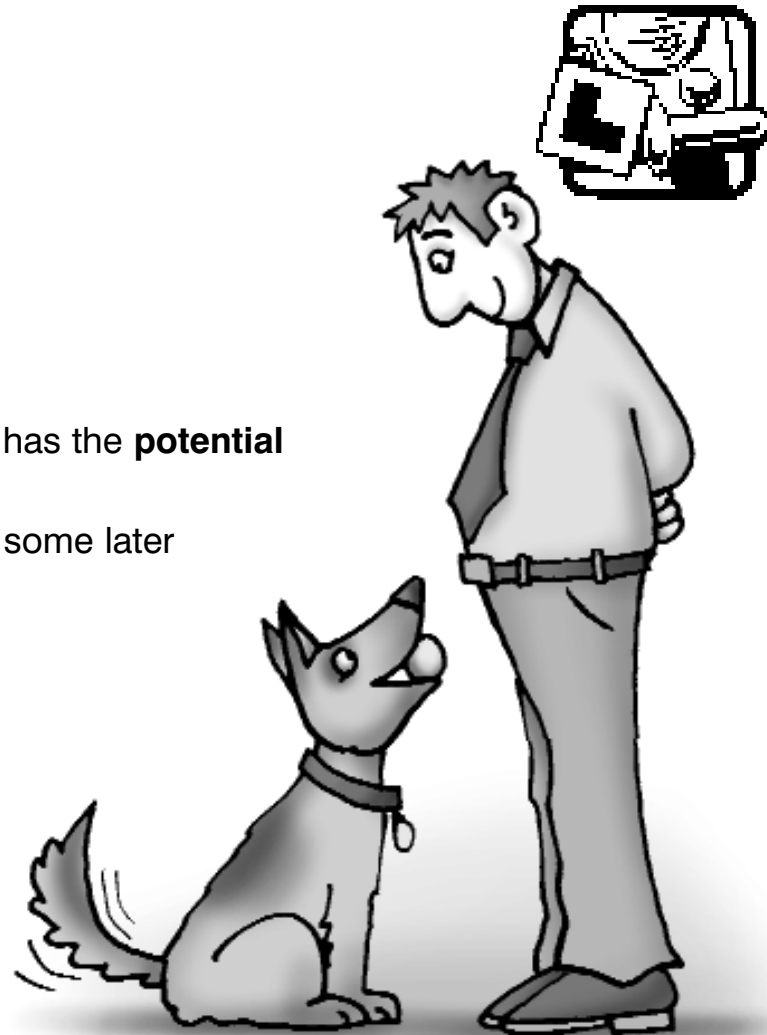
Memory consists of three stages:

- Registration** • You get information
- Retention** • You file information, which has the **potential** to be retrieved
- Retrieval** • You find the information at some later stage when you need it

Retrieval consists of:

- Recall: voluntarily or involuntarily return the information to consciousness
- Recognition: you cannot recall the information but you recognise it when you see it

'Retention' is not the problem, 'retrieval' is.



COMMITTING

MEMORY

RETENTION



You file information, which has the **potential to be retrieved**:

- | | |
|-------------------------------|---|
| Hypnosis | ● Under hypnosis people have recalled <i>forgotten</i> memories perfectly |
| Near death experiences | ● <i>My whole life flashed before my eyes</i>
● Emotion is closely linked to memory
● Highly emotive events are memorable and also stimulate old memories |
| Surprise random recall | ● <i>Déjà vu</i>
● Stimulation of the senses can bring about recall of <i>forgotten</i> memories
● A certain smell or taste brings it all back |
| Dreams | ● Vivid dreams recalling <i>forgotten</i> memories and events |

Turn the page and write down everything you can remember from this page. When you turn back to this page to check, the fact that you recognise it means you have **retention**.

The problem is **retrieval**.

You have experienced **recognition** and involuntary **recall**.

The trick is to master **voluntary recall**.

COMMITTING

MEMORY

HOW IT WORKS

There are three types of memory:

Immediate (sensory) memory – seconds

Limited capacity – seven inputs at once is the average person's limit. This is where input is held until you select what you wish to work with.

Short-term memory – hours or days

This is where you react to new inputs that the immediate memory has selected to work with. Like the immediate memory, this easily reaches overload. To get information into the long-term memory requires repetition of the input in a variety of ways.

(See **Learning Styles** and **Multiple Intelligences**.)

Long-term memory – months or years

This is where you make sense of what you are learning, linking and comparing it to existing learning. The new learning helps shape existing learning, and is then applied.



COMMITTING

MEMORY

SENSORY MEMORY TO LONG-TERM MEMORY

- Learning involves getting information from sensory memory into the long-term memory
- Anything in sensory memory can get in to the long-term memory through **learning**
- Long-term memory has no known storage limit
- When people talk about *memory* they usually mean **retrieval** from long-term memory



COMMITTING

MEMORY

TEST



Read the following list of words, slowly, once. Then cover them up and answer the questions that follow.

HAND
CASE
ME
CAR
BOOK
AND
BRAIN
ME
HERE
TREE
MISSISSIPPI
OF
DOOR
AND
MUSIC
ME
HOME
FILM
WHITE
AND

COMMITTING

MEMORY

TEST



How many of the first five words can you recall?

How many of the last five words can you recall?

Can you recall any words that appeared more than once?

Was there any word that was outstandingly different from the rest?

How many words can you recall that have something to do with what you have already read?

How many of the words from the middle of the list, that you have not already noted above, can you recall?

Read on to check your results.

.....

.....

.....

.....

.....

.....

COMMITTING

MEMORY

LEARNING WITHOUT BREAKS



Primacy: The start of a learning session is more memorable than the middle

First events are more memorable

In the memory test you will have more easily recalled some first words

Recency: The end of a learning session is more memorable than the middle

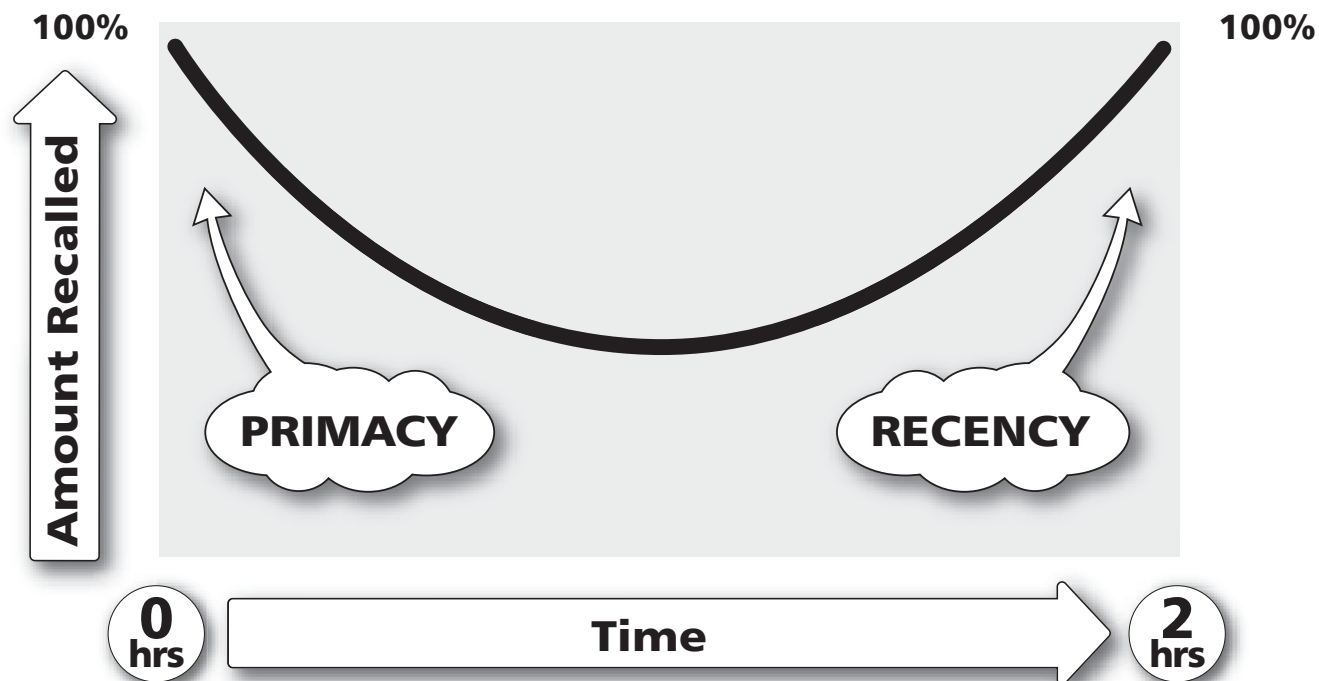
Recent events are more memorable

In the memory test you will have more easily recalled some last words

COMMITTING

MEMORY

LEARNING WITHOUT BREAKS



COMMITTING

MEMORY

LEARNING WITH REGULAR BREAKS



- Keeps recall high, due to the increased effects of primacy and recency
- Increases as the subconscious processes the new information during breaks
- Relieves physical and mental tension

The ideal study time for an adult is 20–30 minutes. Taking a break every half-hour may be disruptive. Instead use a **Brain Gym** exercise. The ideal time for children to study is approximately two minutes in excess of their age. When you take a formal break, perform a change of activity for 5–10 minutes.

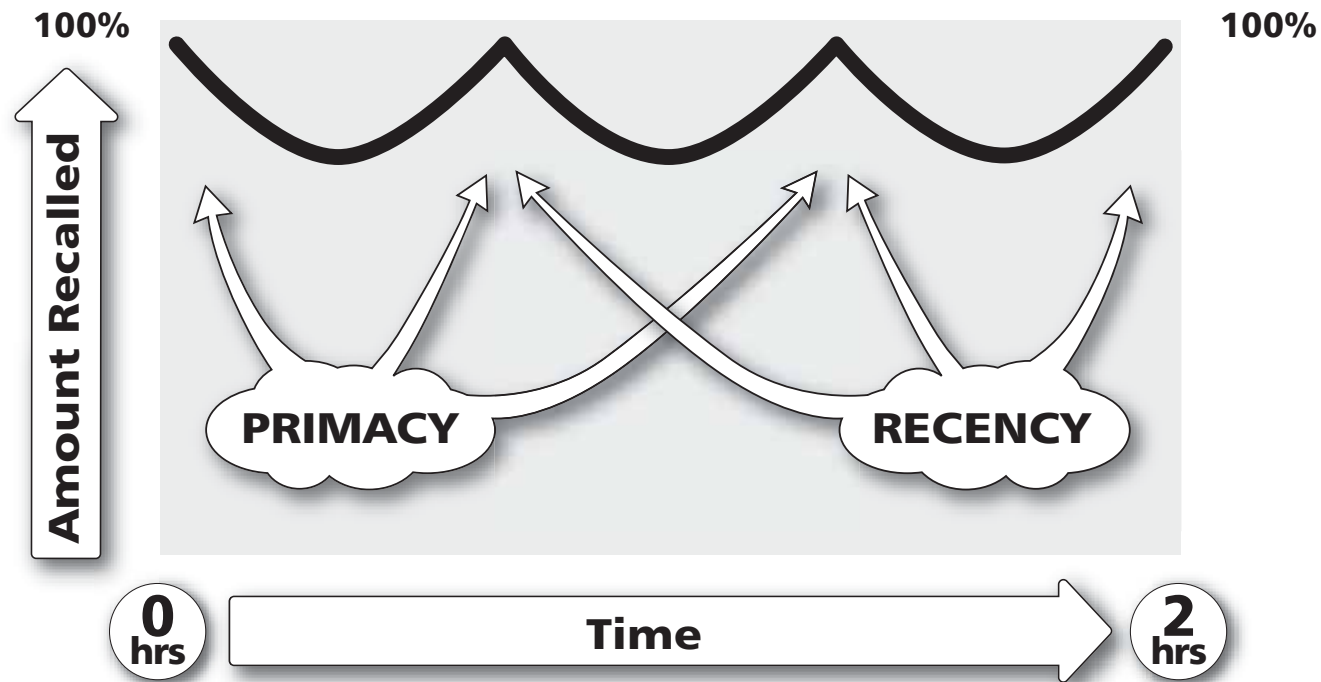
The slow mind is the name given to that part of the brain that continues to process information while you do other things. Your brain makes important connections and finds solutions to problems after you have given it time to digest information. You can help benefit from this process through regular breaks, periods of reflection or by using brain gym exercises.

When you break for lunch, remember to have a light meal and give your body time to digest it before re-starting your learning.

COMMITTING

MEMORY

LEARNING WITH REGULAR BREAKS



COMMITTING

MEMORY

THE EFFECT OF LINKING

- You learn by linking new information to existing knowledge
- Recall occurs when the correct link is made and the information is found
- To aid linking, write down all you know about a subject before beginning any study, read introductory texts, etc

In the memory test you may have recalled book, brain and/or music as the words link to what you will have already read.

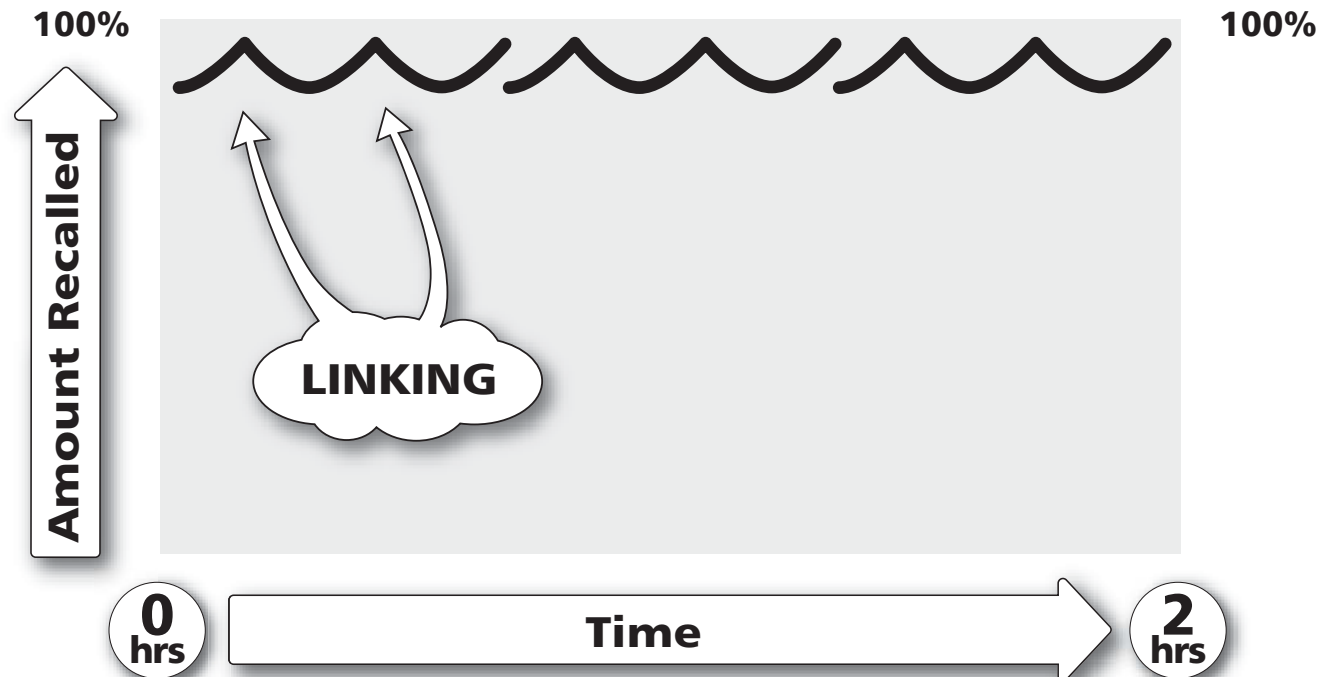
See page 92 for linking techniques.



COMMITTING

MEMORY

THE EFFECT OF LINKING



COMMITTING

MEMORY

THE EFFECT OF OUTSTANDINGNESS

You will recall unusual, outstanding, strange or absurd information more easily (known as the Von Restorff effect).

How often have you forgotten important appointments, names, statistics, etc?
Yet you can still recall vividly something silly that occurred several years ago.

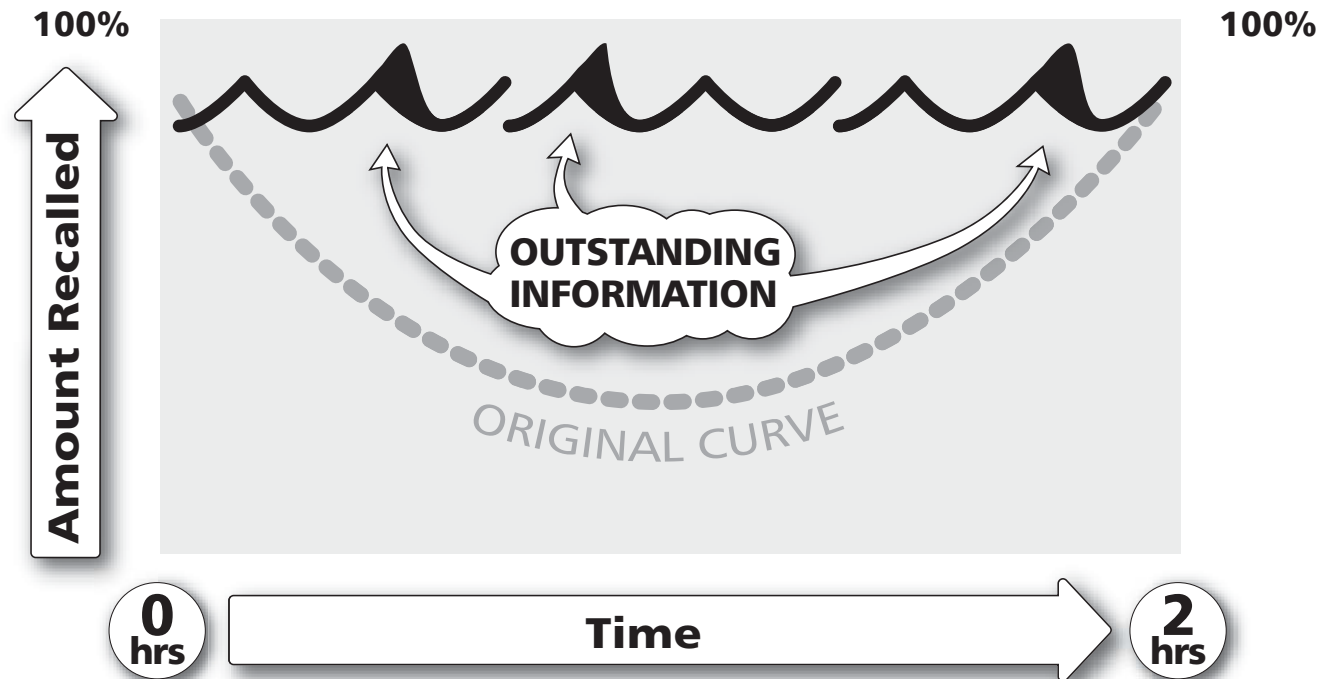
In the memory test you will have recalled the word Mississippi.



COMMITTING

MEMORY

THE EFFECT OF OUTSTANDINGNESS



COMMITTING

MEMORY

LINKING



Link information effectively and creatively by using:

Outstandingness	Exaggerate size, shape, colour, movement and all of the following
Positive images	More likely to recall pleasant images
Humour	Funny events are more positive, outstanding and easier to recall
Sexuality	If you remember one of these seven tips when you turn the page, this will be the one; most people remember well in this area
Sensuality	Make use of all the senses – see, hear, smell, touch and taste
Emotion	Emotive events are more memorable: use love, hate, fear, etc
Personality	There is no substitute for your own images and links; you are more likely to remember what you created Personal events and experiences are more memorable; involve yourself in your images

COMMITTING MUSIC



Certain types of music can match the body's rhythms, heartbeat, brain waves, etc, inducing a state of relaxed alertness, for learning. Dr Georgi Lozanov identified that baroque music, in particular, leaves the mind relaxed and open to learning.

- Classical pieces (1750–1825) and romantic pieces (1820–1900) work best when you are learning new information.
- Baroque pieces (1600–1750) work best when you are reviewing information. The 60-70 beats per minute (the same as the earth's rotation) also induce alpha brain waves (see Brain Waves, page 8)

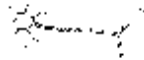
Experiment with different types of music, to see what works best for you. If you find it distracting – don't use it.

COMMITTING

MIND MAPS

BENEFITS

Mirrors how the brain looks and works



Stimulates both right and left hemispheres of the brain



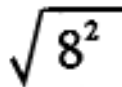
Makes use of linking



Makes use of different learning styles



Makes use of different intelligences



Visual excitement aids memory



Your ideas are easier to recall



Saves time, only recording and reviewing key words

Easy to review, recreate from memory and check against original

Numerous other uses eg making a presentation, project planning, writing reports, creativity and brainstorming sessions, taking minutes

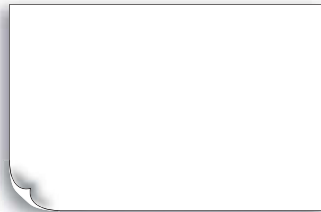
The Mind Map® created and designed by Tony Buzan. Learner's Pocketbook Mind Mapping officially approved by Tony Buzan.

COMMITTING

MIND MAP MAXIMS



Paper



Good quality to stimulate all the senses
Blank – nothing to confine your creativity
Landscape – maximum space
Only use one side – easier to read

Central Image



Has more associations than a word and aids memory
Central image attracts the eye
Size 5 x 5 cm (2" x 2"), open no frame
Use at least four colours, appealing – encourages learning

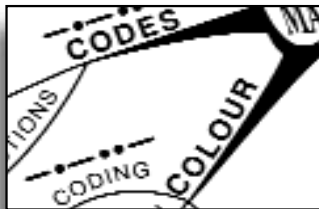
If you must use a word, use **DIMENSION** and colour

COMMITTING

MIND MAP MAXIMS



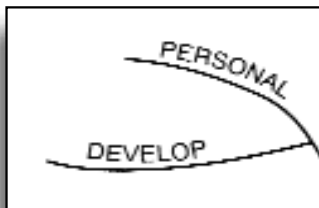
Branches



Main branches thicker, symbolise importance
Curved lines – stimulate visual interest
Length of line = length of word

STIMULATE

Words



Only record key words
PRINT WORDS ON LINE
Main branch word – UPPER CASE
Secondary branches – lower case
Vary size relative to **IMPORTANCE**
Use for

one word per line

EMPHASIS

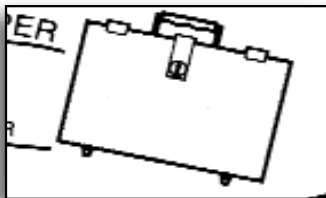
STRESS

COMMITTING

MIND MAP MAXIMS



Images



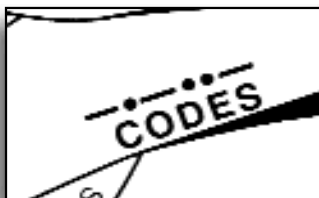
Stimulate the right brain – aid memory
Have more associations than words
Attract the eye, encourage learning
Eye takes in images faster than words
Where possible replace words with pictures

Colour

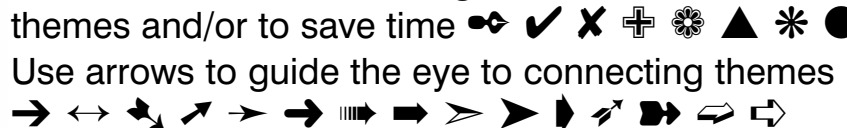


Stimulates the right brain – aids memory
Use increases short-term memory by 75% and long-term by 25%
Use one colour per main branch theme
Use for coding, enables faster access to information

Codes



Use codes for instant linking of information, recurring themes and/or to save time
Use arrows to guide the eye to connecting themes



COMMITTING

MIND MAP MAXIMS



Spacing



Leave space
for CLARITY and ADD + ITIONS

Personal style



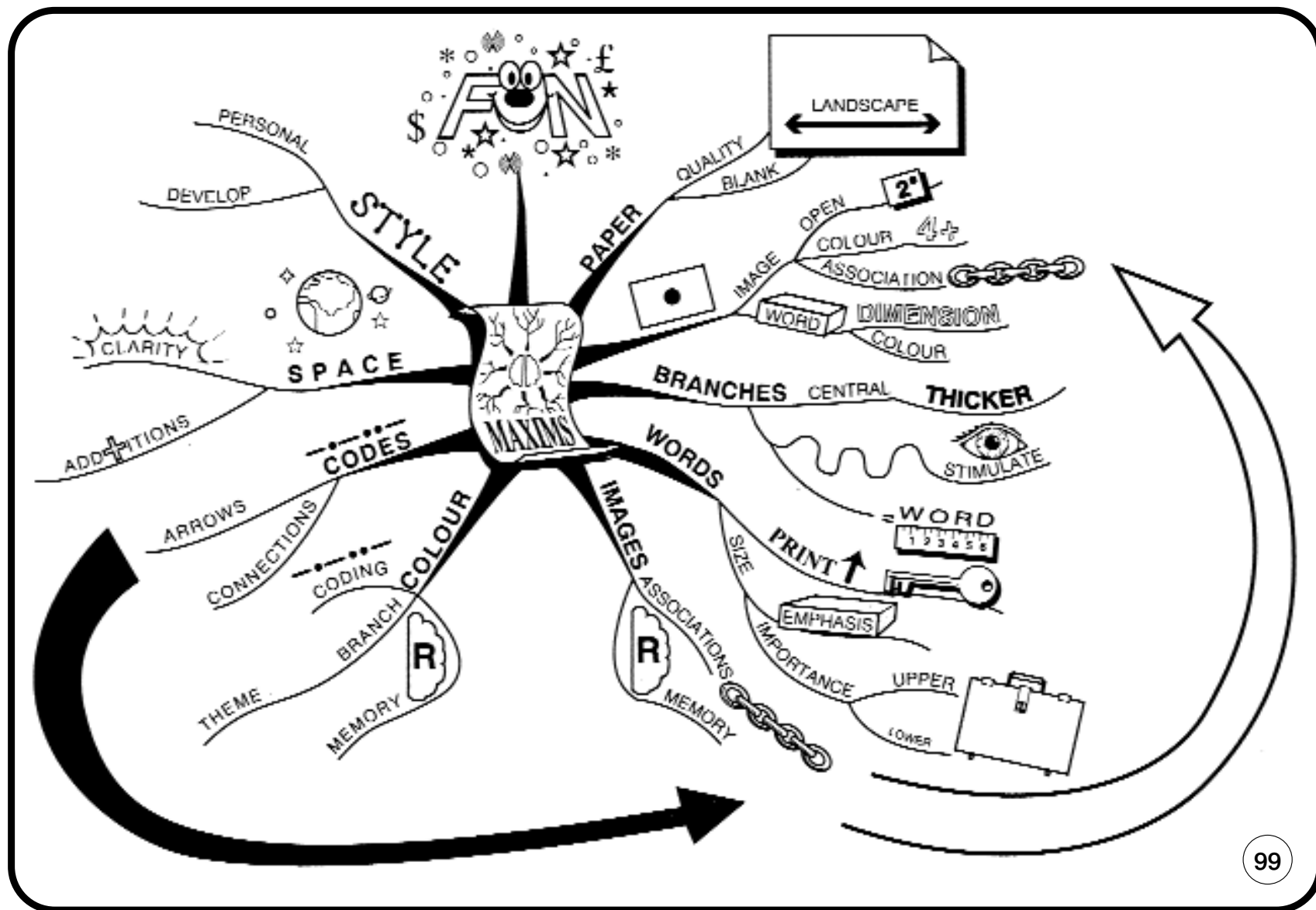
Learning style is personal
Personal events are more memorable
Develop a personal style for your mind maps

Have fun

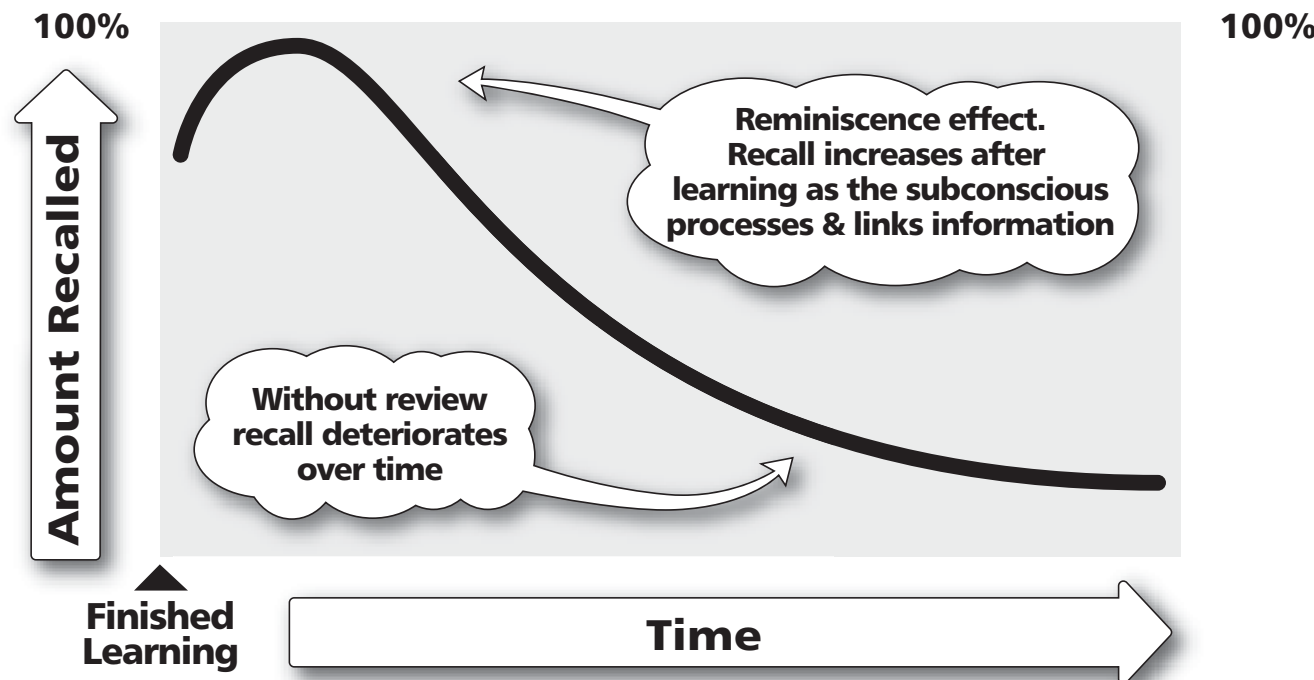


When learning is fun you learn faster and recall more
You are more likely to repeat the experience
It takes 72 muscles to frown and only
14 to smile

**Relax and
enjoy learning**



COMMITTING REVIEW



COMMITTING

REVIEW

WHY & HOW

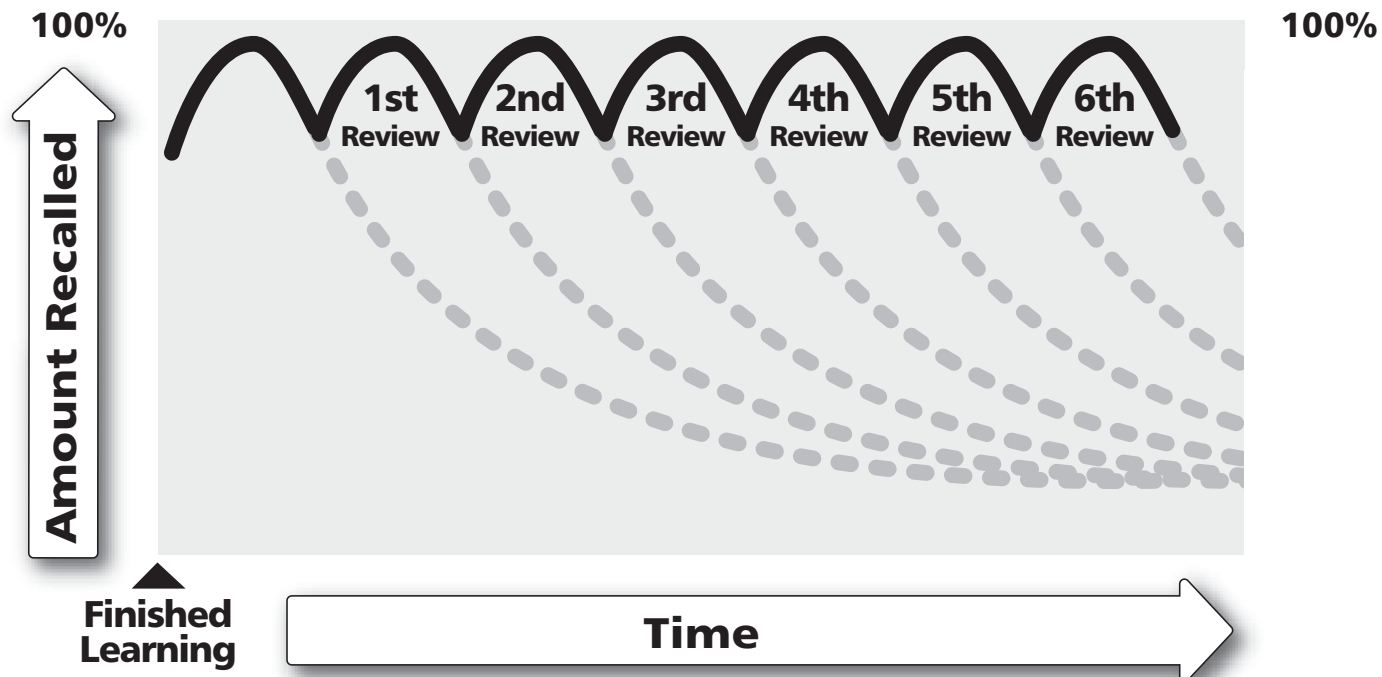


- Once you have learned something, it is easier to keep that knowledge fresh than to start again
- Do not let what you have learned go to waste
- **Reviewing** embeds that knowledge in your long-term memory
- Repetition is not enough – **use a different learning method**
 - review mind maps by re-creating
 - then review only what you want to remember
 - key facts only
 - not everything you have read, heard, seen or done

COMMITTING

REVIEW

WHEN



COMMITTING

REVIEW

WHEN



REVIEW

WHEN

HOW LONG

1st review

10–30 minutes after
a learning event

For 5 minutes

2nd review

1 day after

For 5 minutes

3rd review

1 week after

For 5 minutes

4th review

1 month after

For 3 minutes

5th review

3 months after

For 3 minutes

6th review

6 months after

For 3 minutes

S t o r e d i n l o n g - t e r m m e m o r y

COMMITTING MENTORING



- A mentor is an individual skilled in the area you are learning
- They should be supportive and provide further information
- Watch them, talk to them, use their ideas and methods
- There should be a two way contract which is mutually beneficial to both parties – even if it is only buying lunch, while you pick their brains
- Return the favour to someone else by being a mentor yourself





REFLECTING

REFLECTING

TEST YOURSELF



Have you learned?

- Before testing yourself, set a standard and keep it
- Speed is not important – getting it right is
- Use, or create, mock exams
- Use mental rehearsal/visualisation
- Re-create mind maps from memory
- Teach others (your learning set) – this really proves if you know your stuff
- If you are learning to learn – learn something, put it to the test

REFLECTING

TEST YOURSELF



If you learn to drive, you get in a car.

If you learn to swim, you jump in the water. So:

- If you are learning how to mind map – mind map books, TV, documentaries, etc
- If you are learning how to write – put pen to paper
- If you are learning French – *parlez francais*
- If you are learning Chinese cookery – buy a wok and start using it
- If you are learning about computers – sit in front of one and use it
- If you are learning how to deliver speeches – stand up and talk
- If you are learning book keeping – get your calculator out
- If you are learning how to learn – learn something, use the techniques in this book

No child has learned to talk only by watching others and reading about it; sooner or later they open their mouth.

REFLECTING

LEARNING SET



Test yourselves:

- Present your learning to the set
- Test the set
 - set tests
 - run quizzes
 - have question and answer sessions
- Compete against other learning sets

Ask yourselves:

What worked well?

What could we do better next time?

Points to consider:

- Contract
- Competition
- Support
- Mix of styles
- Openness
- Quality of work
- Roles
- Benefits

REFLECTING

IMPROVING



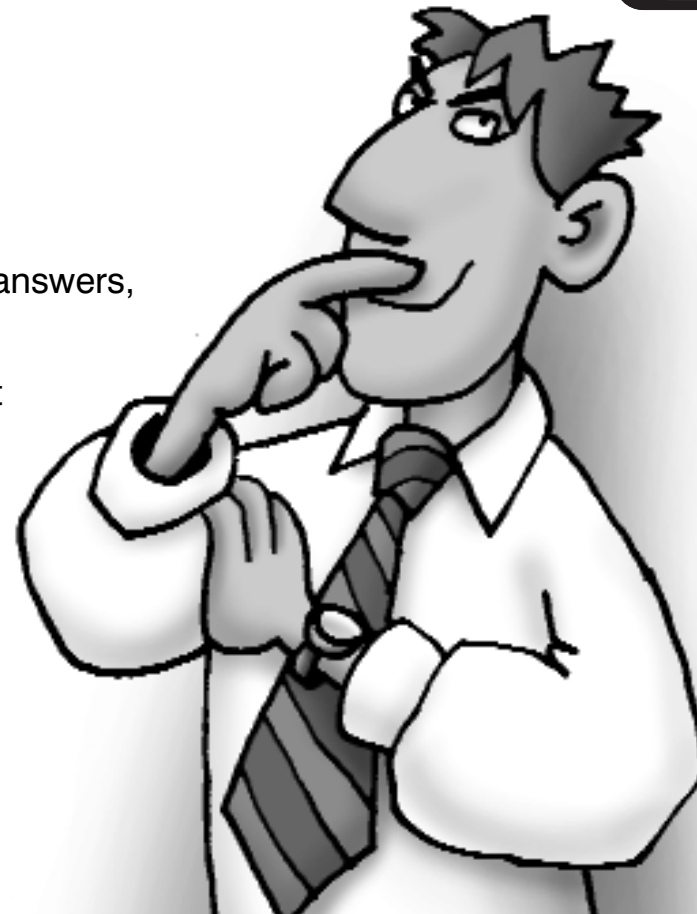
What worked well?

What could I do better next time?

If you ask these questions and act on the answers, you will develop your learning ability.

It is as important to understand and repeat what did work well as it is to improve on anything that did not.

Remember – you are responsible for your own learning.



REFLECTING

REWARD



You should recognise and reward your successes.

- It encourages positive emotions about learning
- Ensures learning is fun, rewarding, enjoyable and worth doing again

Incentives should be given along the way, but, in particular, at the end of a learning event. The reward is not: having the knowledge, passing the exam, getting the qualification, achieving the promotion or the feeling that any of this gives you. Reward is:

- Eating out at a special restaurant
- Having a weekend break
- Going to the theatre
- Doing something unusual:
 - bungee jumping
 - driving a racing car
 - water ski-ing, etc
- Sharing your success with others

**How will you
reward yourself for
learning how to learn?**



MEMORY TECHNIQUES

MEMORY TECHNIQUES

METHODS



Principles – Principles are easier to recall than individual examples, eg *i before e, except after c, or when sounded like a, as in neighbour or weigh*

Rhyme – Makes use of musical intelligence, eg *30 days hath September ...*

Acronym – Forming a word from the first letter(s) of other words, eg:

H	uron	
O	ntario	
M	ichigan	The great lakes
E	rie	
S	uperior	

Mnemonic – Generic term for any memory aid, eg ‘**E**very **G**ood **B**oy **D**eserves **F**avour’ **EGBDF**, the musical notes on the line of the stave

Storytelling – Although it may be many years later, the moral of a story sticks with us, eg *The Tortoise and the Hare*.

MEMORY TECHNIQUES

NUMBERS



Numbers are traditionally hard to remember. They lack feeling, image, motion, humour and are, in comparison to a word, meaningless.

Use the following method to convert numbers to words and pictures, therefore making them easier to remember.

Number	Code	Remember
0	S, Z	first sound of the word zero
1	T, D	both t and d have one downstroke
2	N	n has two downstrokes
3	M	m has three downstrokes
4	R	final sound of the word 'four'
5	L	roman numeral for 50, make L with your hand (5 digits)
6	J, CH, SH soft G	j turned around is like the number 6
7	K, hard C, hard G	k formed from two 7s, one reversed
8	F, V	f and 8 both have two loops, one above the other
9	P, B	p reversed is 9



MEMORY TECHNIQUES

NUMBERS IN PRACTICE



This system can be used to remember any sequence of numbers, eg:

123 = DNM = **Denim**
921210 = PNDNTS = **Pound notes**

number →

word →

picture →



Vowels and the letter W, H and Y (WHY) have no meaning under this system and are used to complete words.

What about the following?

number →

word →

picture →

Answer at bottom of page

54226 = ? ? ? ? ? = ?

Practise the system regularly, break numbers down into words for speed.

Based on the work of Stanislaus Mink Von Wennsshein 1648 and Dr Richard Grey 1730

MEMORY TECHNIQUES

NUMBERS



The number system covered is worth learning; why not try the following method too, eg:

Pi = 0.318310	0.	3	1	8	3	10
		▽	▽	▽	▽	▽
	Can	I	remember	the	reciprocal	

The letters in each word total the number you want to remember.

What about our earlier number 5 4 2 2 6 ? Try to come up with a memorable phrase.

Suggestion at bottom of page.

MEMORY TECHNIQUES

DATES



Try using the system covered earlier, on page 113.

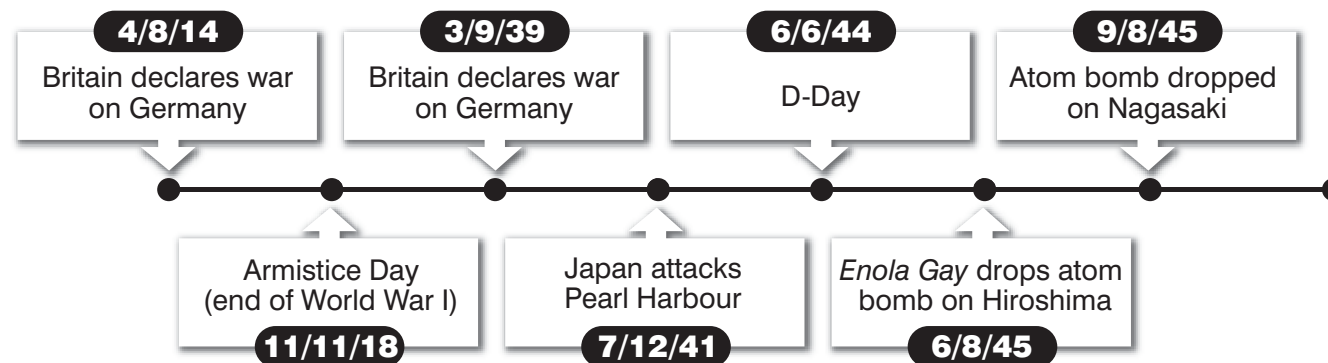
Wall Street Crash 1929 2 = N, 9 = P

N a P = Picture yourself driving a car, having a nap and crashing into a wall.

Chamberlain declares war on Germany 3/9/1939 3 = M, 9 = D

M a D M a D = Picture a mad and angry Chamberlain fighting a mad and angry Hitler.

And/or try a time line:



MEMORY TECHNIQUES

PEG SYSTEMS



This is a method used for remembering lists and sequences. You use a memorised sequence of hooks, on to which you link new information. (For effective links, see **Linking**, page 88.) Make your peg system personal by creating your own words and images. Personal ideas are easier to recall. (The number system below could be used as a peg system, giving you thousands of pegs on to which you can link new information.)

Number System

1	=	D	=	Day
2	=	N	=	Noah
3	=	M	=	May
4	=	R	=	Ray
5	=	L	=	Lay
6	=	J (ch, sh, soft g)	=	Jay
7	=	K	=	Key
8	=	F (l, v)	=	Fee
9	=	B (l, p)	=	Bay
10	=	DZ	=	Daze

Number Rhyme

1	=	Gun	
2	=	Chew	
3	=	Tree	3 
4	=	Door	4 
5	=	Hive	
6	=	Bricks	5 
7	=	Heaven	
8	=	Gate	
9	=	Wine	9 
10	=	Hen	10 

MEMORY TECHNIQUES

PEG SYSTEMS

NUMBER SHAPE

1 = Pen



6 = Yo-Yo



2 = Swan



7 = Boomerang



3 = Hills



8 = Egg-timer



4 = Boat



9 = Flag



5 = Hook



10 = Bat & ball



MEMORY TECHNIQUES

PEG SYSTEMS

ALPHABET SYSTEM



The peg word must start with the sound of the letter, eg:

A = **A**ce not **A**pple, C - **S**ea not **C**at.

A = Ace

B = Bee

C = Sea

D = Deed

E = Easel

F = Effluent

G = Gee Gee

H = H-Bomb

I = Eye

J = Jay

K = Cake

L = Elbow

M = Empire

N = Enema

O = Obese

P = Pea

Q = Queue

R = Arm

S = Eskimo

T = Tea

U = Yew

V = Venus

W = WC

X = X-ray

Y = Y-fronts

Z = Z car

MEMORY TECHNIQUES

PEG SYSTEMS



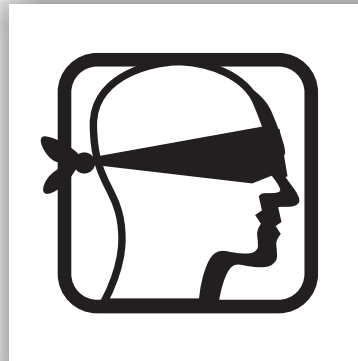
Days of the Week

Sunday	–	sun
Monday	–	money
Tuesday	–	chew
Wednesday	–	wedding
Thursday	–	thirsty
Friday	–	fry
Saturday	–	saturn

Months of the year

January	–	New Year
February	–	Brewery
March	–	Marching
April	–	Ape
May	–	Maypole
June	–	Bride
July	–	Jewel
August	–	Gust
September	–	Sceptre
October	–	Octopus
November	–	Fireworks
December	–	Christmas

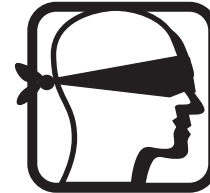
Use your own *personal* images – more memorable.



BLOCKS TO LEARNING

BLOCKS TO LEARNING

OVERCOMING BLOCKS TO LEARNING



Psychological: Wrong state of mind, negative attitude to learning, fear

Use relaxation techniques to prepare for learning and exercise regularly

Have a clear focus on why you are learning

Use positive affirmations and visualisation to reinforce success

Reward yourself, make learning fun

Use your learning set or mentor to support your fears and praise your success

Physiological: Poor environment, distractions, too comfortable and relaxed

Prepare your environment for learning

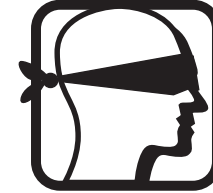
Avoid distractions, such as noise, visual distractions and comfortable armchairs

Learning style: Not knowing how you learn best

Find your strengths and use them to your advantage

BLOCKS TO LEARNING

OVERCOMING BLOCKS TO LEARNING



Time: **Not having the time, learning solidly for hours, burning the midnight oil – especially prior to an exam**

Plan out your learning (use a learning plan)

Prior to an exam, relax – don't cram. With a learning plan all your learning will have been completed

Subject: **Lack of interest, lack of understanding**

Have a clear focus on why you are learning

Reward yourself

Read simple introductory texts at first

Prepare your questions (the six W's) and get involved

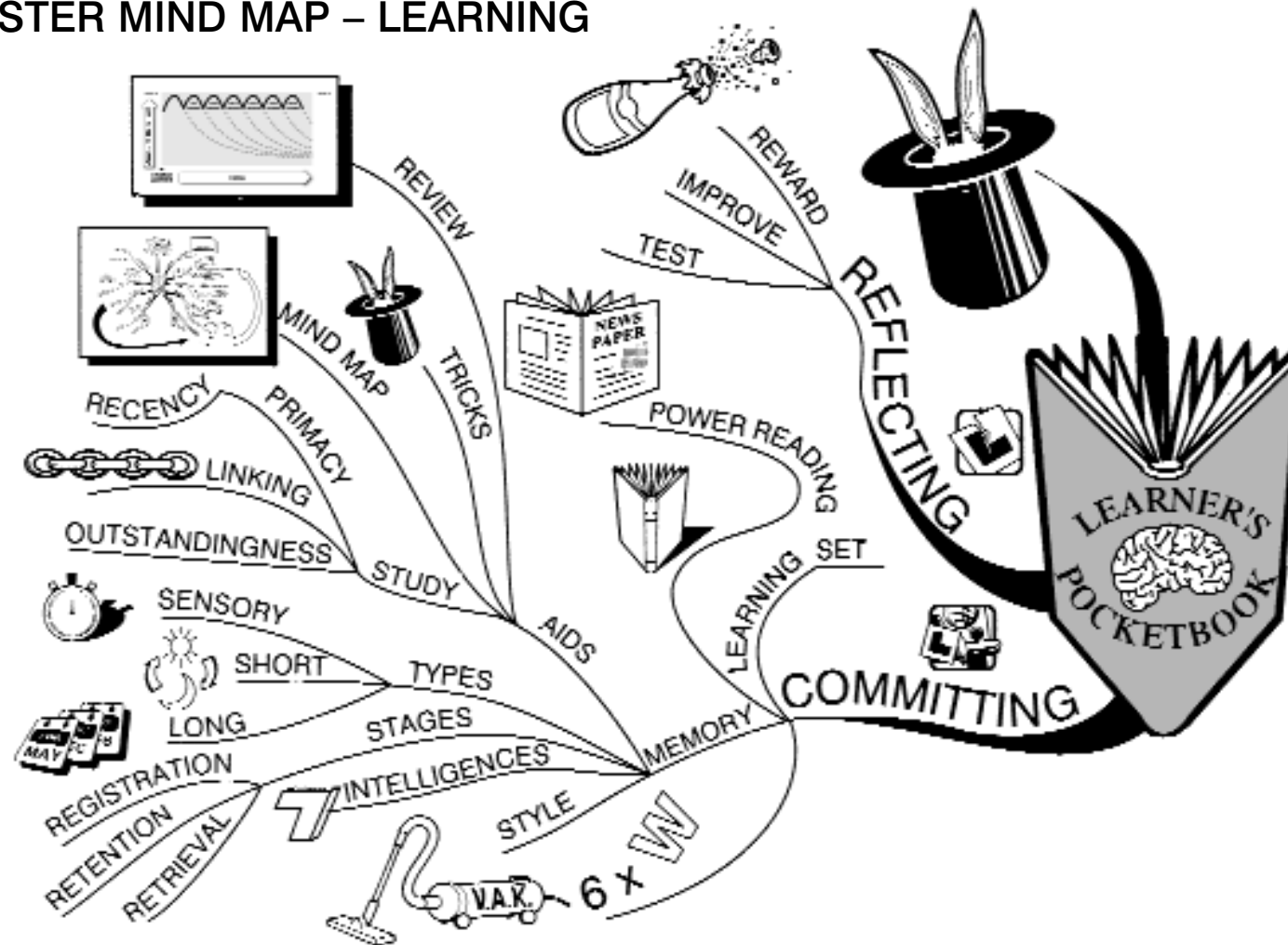
Experience: **Previous bad experiences of learning**

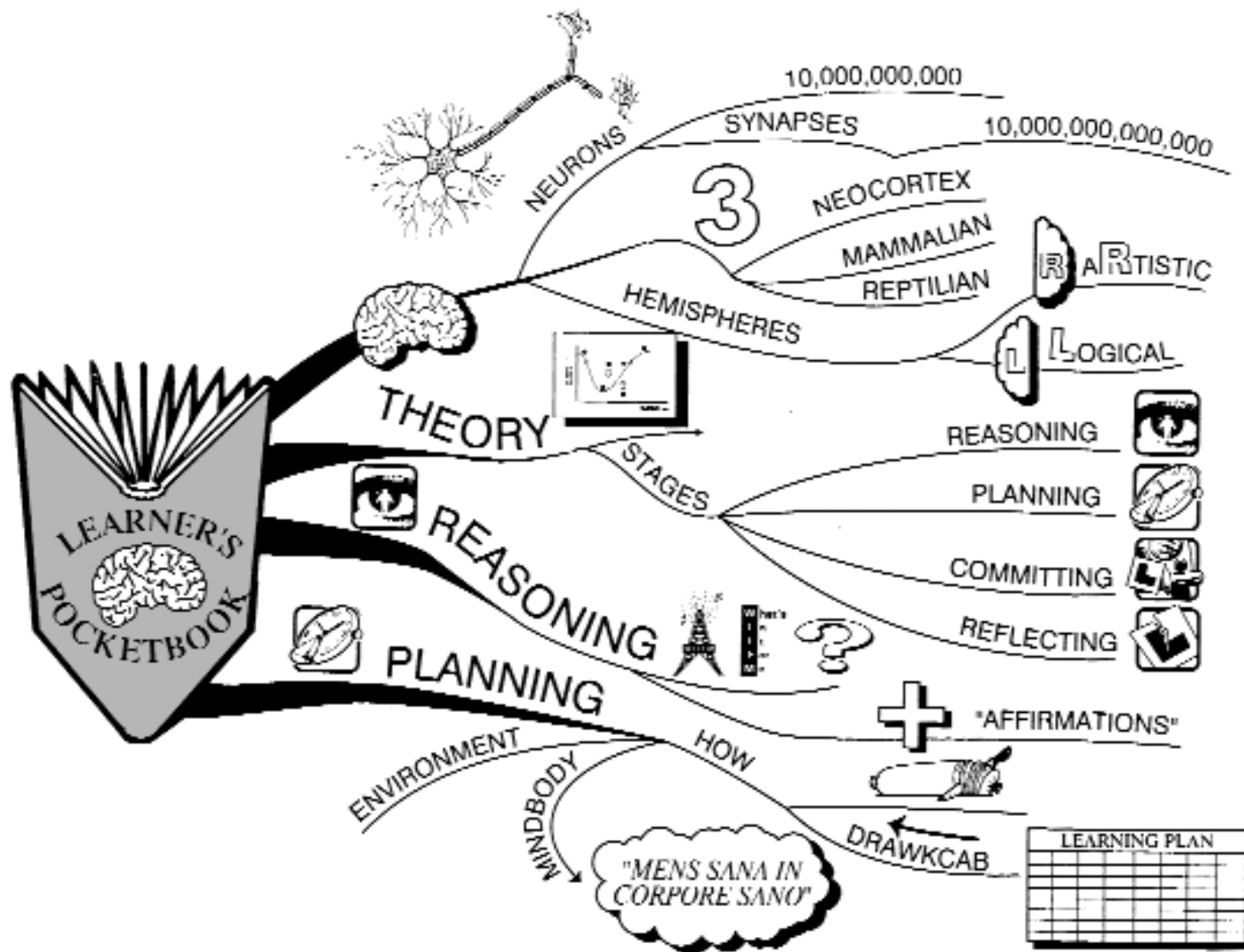
All of the above may have caused this experience

Is it really as bad as you perceive or have you created it?

You cannot change the past, but now is the time to move on

MASTER MIND MAP – LEARNING





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Paul runs his own training and consultancy business. His clients include KPMG, Bank of Scotland, IBM, Prudential and BusinessLink. In his role as a training consultant, Paul works to develop a wide range of skills and knowledge to help both people and companies maximise their potential.

Prior to running his own company, Paul worked at Allied Dunbar where he had responsibility for the development of head office personnel and, later, the salesforce. Paul is author of several in-house training manuals, 'The Personal Success Pocketbook' and co-author of 'The Financial Adviser's Guide'.



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